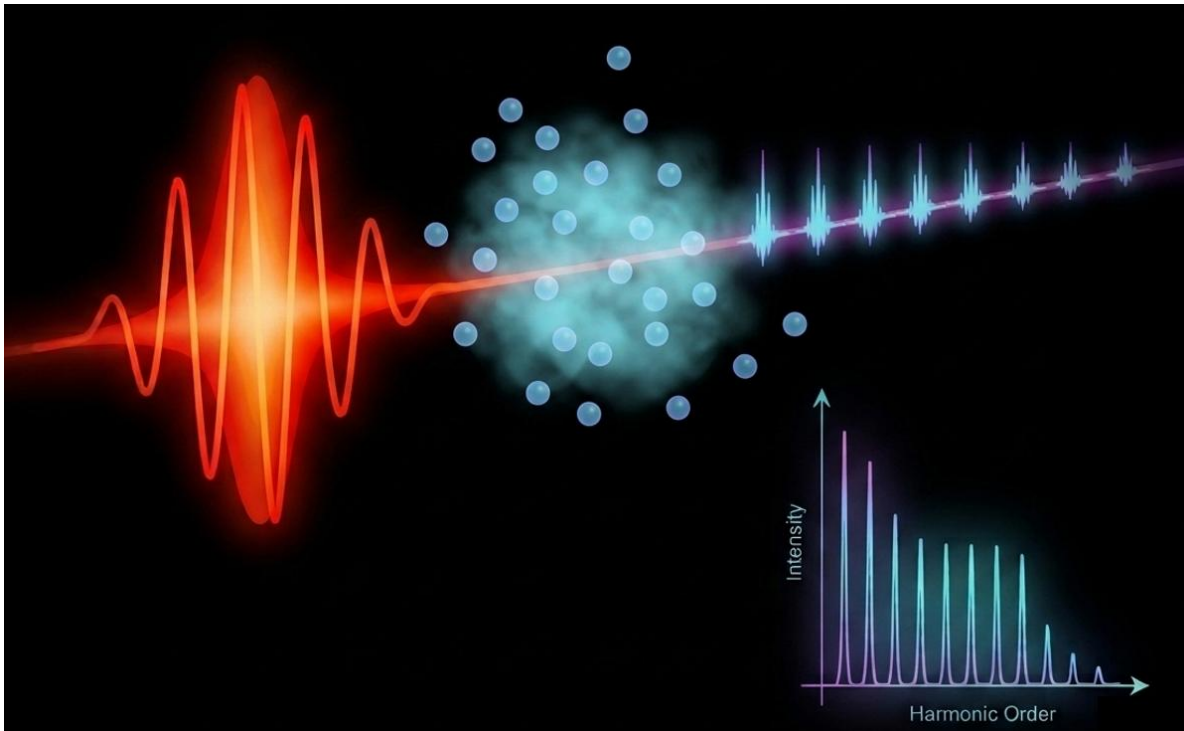


High harmonic generation and ultrafast electron dynamics simulations



IMPRS UFAST focus course

17th, 18th , 24th, 25th August
9:00-12:00h

Alba de las Heras
Theory department

UNTERSTÜTZT VON / SUPPORTED BY



Alexander von
HUMBOLDT
STIFTUNG



mpsd
Max-Planck-Institut für
Struktur und Dynamik der Materie

Outline

Session 1: Fundamentals of Strong-Field Physics

Session 2: Microscopic modeling of laser-driven electron dynamics and spectroscopic signatures in high harmonic generation

Session 3: Macroscopic picture of high harmonic generation and control of the properties of structured attosecond pulses

Session 4: Solid-state attosecond physics

Contents of Session 3:

Macroscopic picture of high harmonic generation

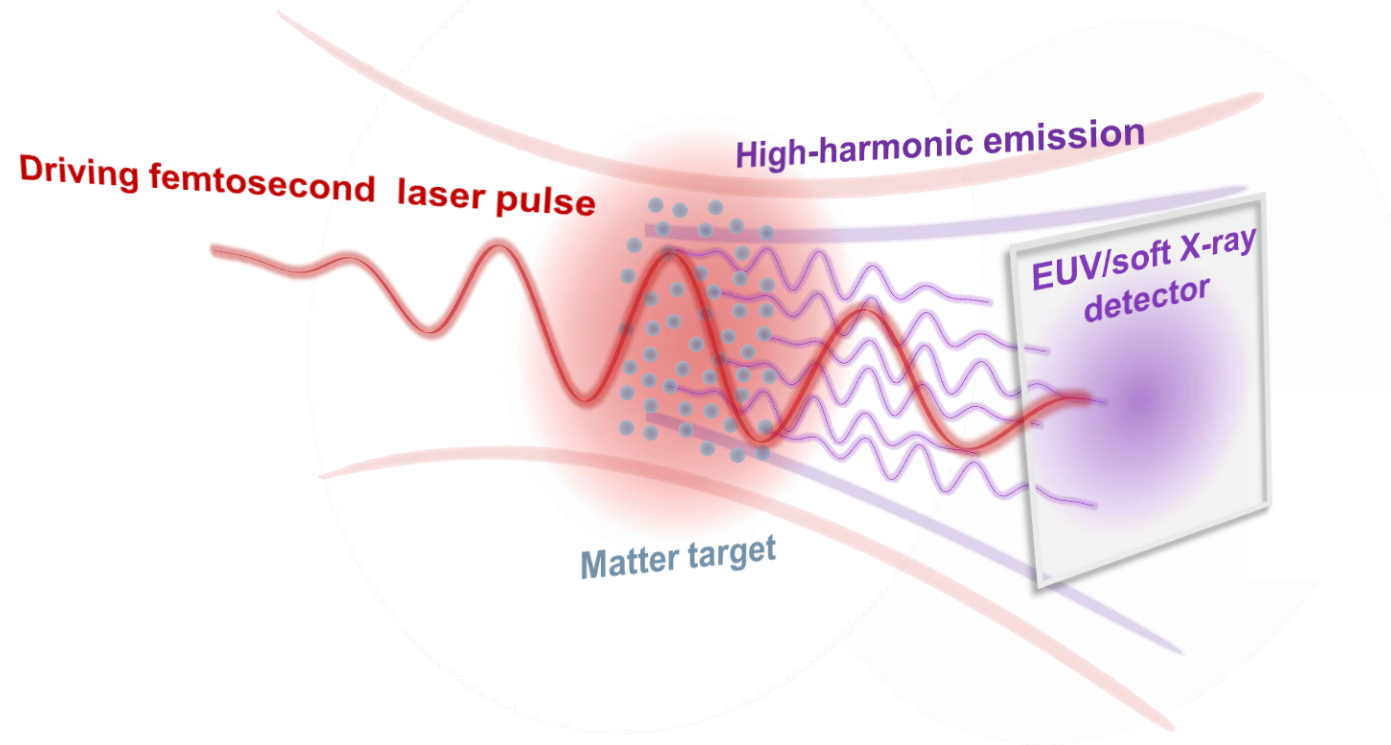
Dynamical symmetries in laser-matter interactions

Overview of chiral light

Tailoring extreme ultraviolet radiation via high harmonic generation

Circular and helical dichroism for chiral detection

Macroscopic picture of high harmonic generation



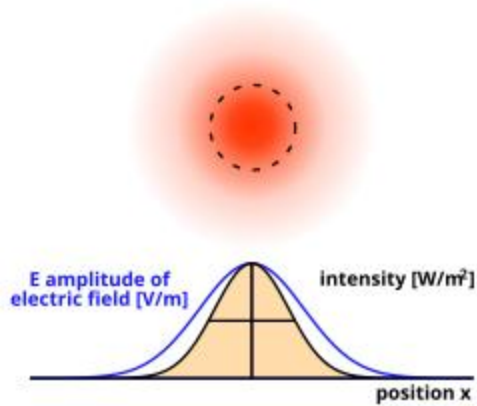
Macroscopic effects in high harmonic generation

To model the macroscopic physics of high harmonic generation, we need to consider:

- Macroscopic laser beams (spatial dependencies become important, different collinear/noncollinear configuration schemes, etc.)
- Optical response (absorption and reflection) at the medium interface.
- Propagation of the laser beam in the medium.
- Free-space propagation to far-field detectors.

Gaussian laser beam

Transverse beam profile:



Wavenumber: $k = 2\pi n/\lambda$

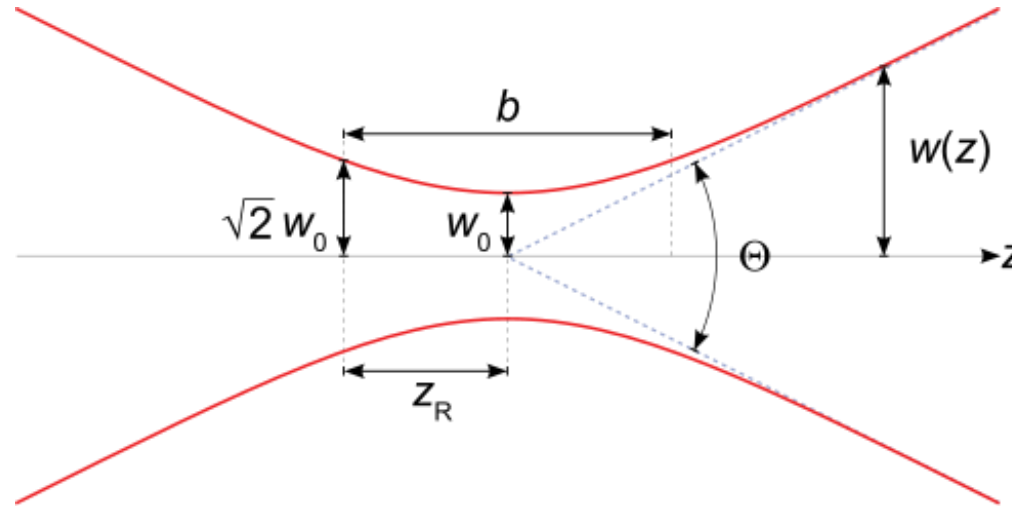
Rayleigh length: $z_R = \frac{\pi w_0^2 n}{\lambda}$

Radius of curvature: $R(z) = z \left[1 + \left(\frac{z_R}{z} \right)^2 \right]$

Gouy phase: $\psi(z) = \arctan\left(\frac{z}{z_R}\right)$

$$\mathbf{E}(r, z) = E_0 \hat{\mathbf{x}} \frac{w_0}{w(z)} \exp\left(\frac{-r^2}{w(z)^2}\right) \exp\left(-i \left(kz + k \frac{r^2}{2R(z)} - \psi(z) \right)\right)$$

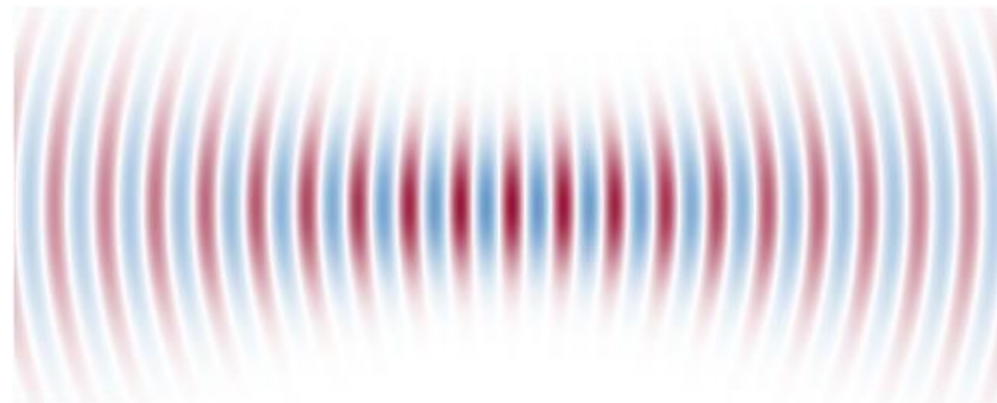
Longitudinal beam profile:



Beam width:

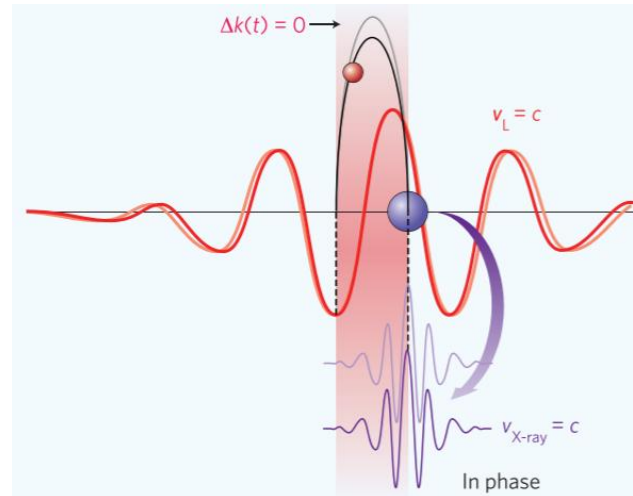
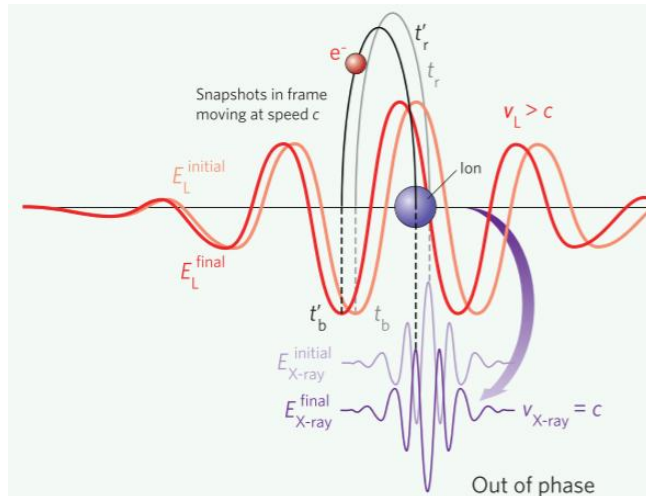
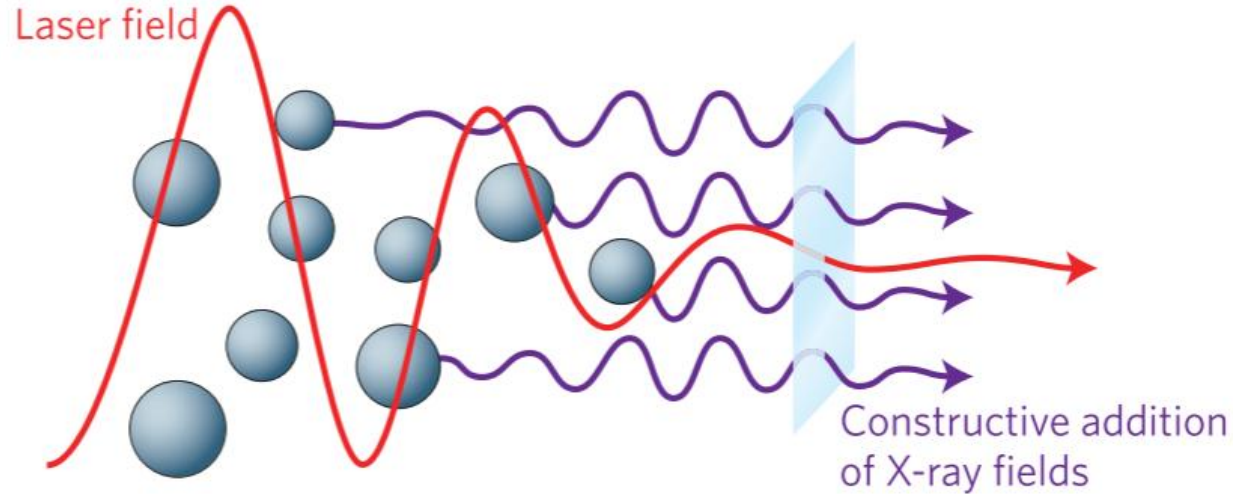
$$w(z) = w_0 \sqrt{1 + \left(\frac{z}{z_R} \right)^2}$$

Beam waist: $w_0 = w(0)$



Phase matching high harmonic emission

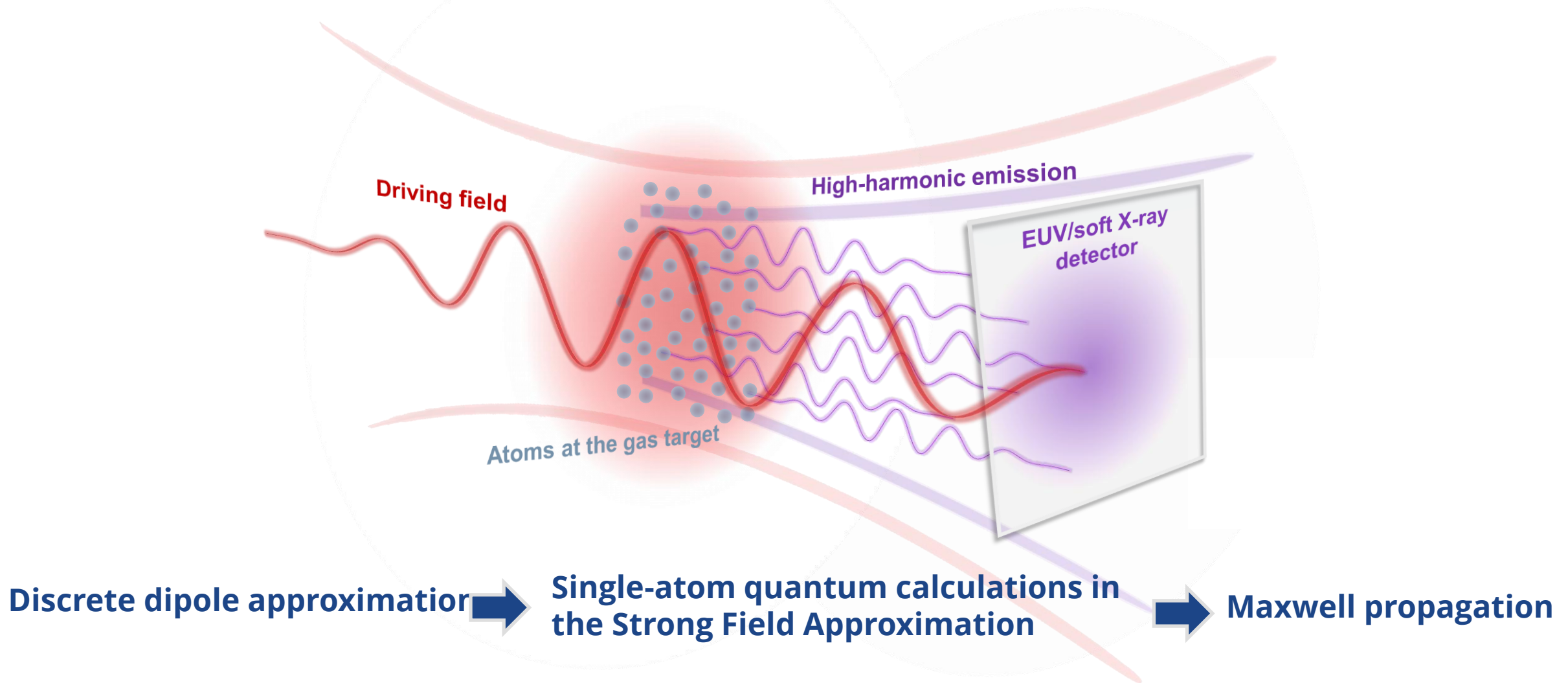
Macroscopic phase matching $\Delta k_q = k_q - qk = 0$



The HHG efficiency is maximized when there is constructive interference of the harmonic radiation from different spatial points. This occurs when the phase-matching condition is fulfilled.

To achieve optimum phase-matching conditions, one needs to adjust the peak intensity, the gas pressure, the spatial distribution of the driving beam, the laser focusing configuration, and the characteristics of the gas target.

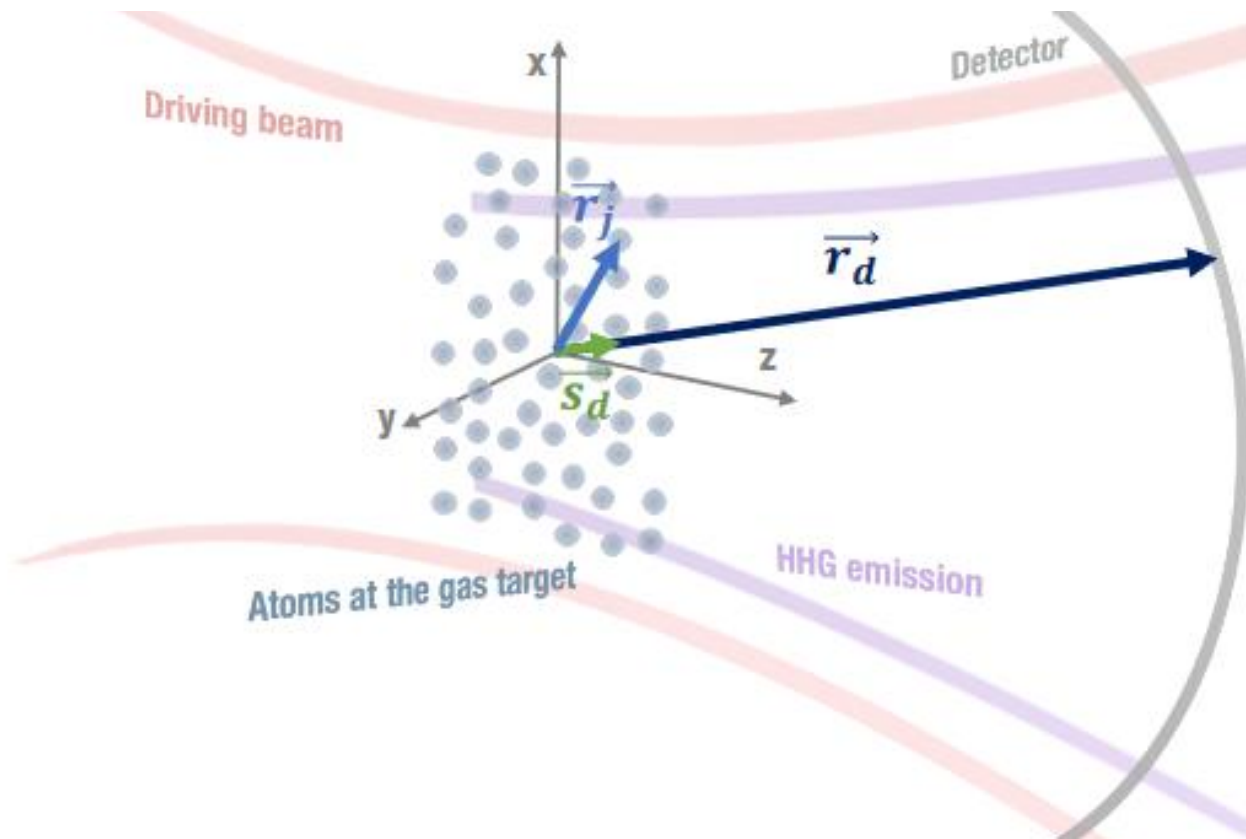
Propagation model of HHG based on discrete dipole emitters



Free-space propagator to far-field detectors:

Electric field of the elementary dipoles at the far-field detector:

$$\mathbf{E}_j(\mathbf{r}_d, t) = \frac{q_j \mathbf{s}_d}{4\pi\epsilon_0 c^2 |\mathbf{r}_d - \mathbf{r}_j|} \times \left[\mathbf{s}_d \times \mathbf{a}_j \left(t - \frac{|\mathbf{r}_d - \mathbf{r}_j|}{c} \right) \right]$$



Total electric field at the far-field detector:

$$\mathbf{E}_T(\mathbf{r}, t) = \mathbf{E}(\mathbf{r}, t) + \sum_{j=1}^n \mathbf{E}_j(\mathbf{r}, t).$$

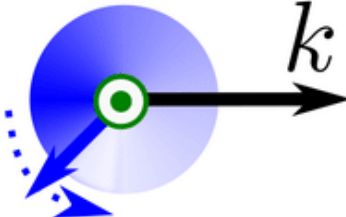
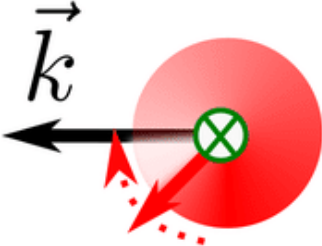
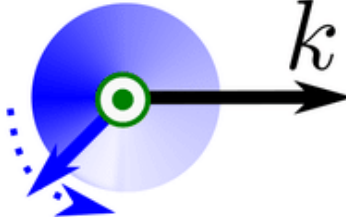
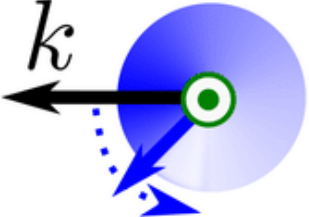
Dynamical symmetries in laser-matter interactions

(important in both microscopic and macroscopic pictures)

Dynamical symmetries in laser-matter interactions

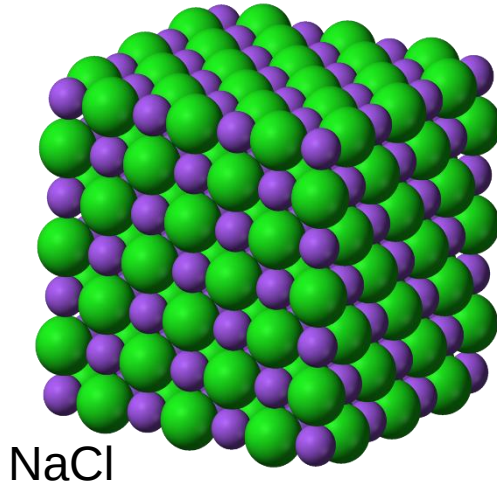
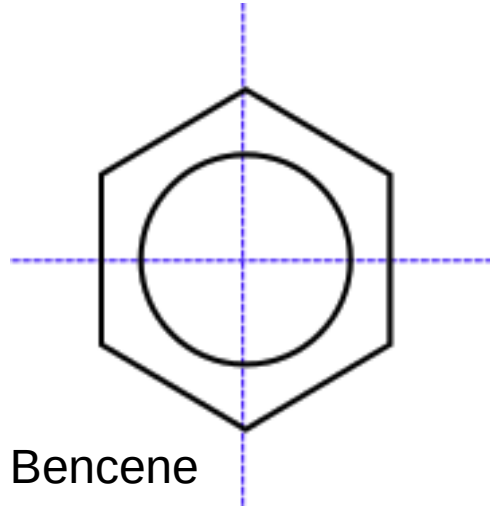
- **Dynamical symmetries are joint space-time symmetries shared by the driving laser field and the target system's Hamiltonian. Thus, the recipe is:**
 - 1. Identify the symmetry of the matter system.**
 - 2. Identify the dynamical symmetries of the driving laser field.**
 - 3. Identify the dynamical symmetries of the joint system (laser driver + matter). This will govern the optical response.**

Time-reversal symmetry

	Forward propagation	Time reversal	Reversed propagation
$t \leftrightarrow -t$ Time reversal symmetry			
$t \nleftrightarrow -t$ Broken time reversal symmetry			
	Spin-momentum locking		
	Broken spin-momentum locking		

Inversion symmetry

Inversion symmetry: $(x, y, z) \leftrightarrow (-x, -y, -z)$



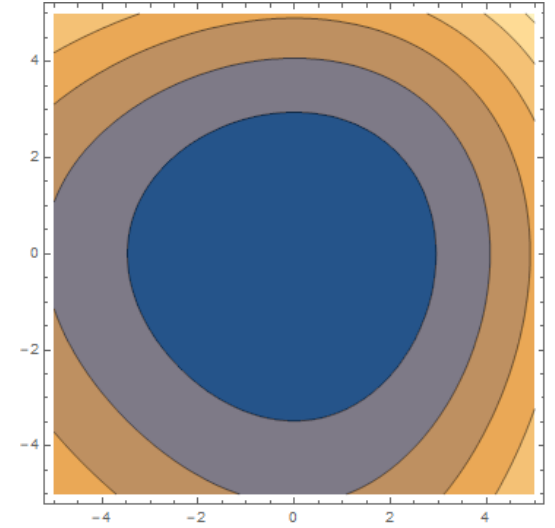
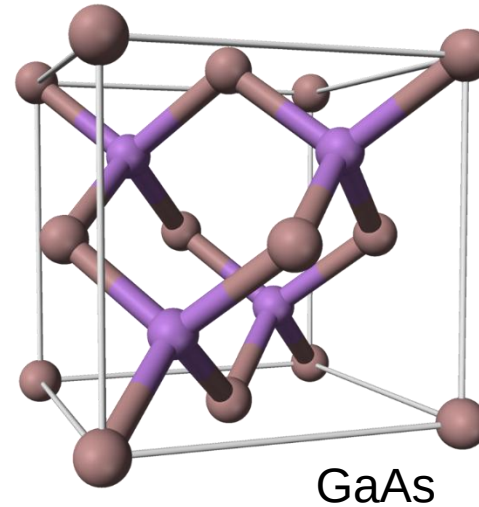
Only odd harmonics
when using a monochromatic driving field

$$d(t + T/2) = -d(t)$$
$$(-1)^{n_H} = -1$$

Parity conservation

No inversion symmetry:

$(x, y, z) \leftrightarrow (-x, -y, -z)$



Odd and even harmonics

Inversion symmetry in laser-matter interactions

Inversion symmetry: $(x, y, z) \leftrightarrow (-x, -y, -z)$

Only odd harmonics

when using a monochromatic driving field

$$d(t + T/2) = -d(t)$$

$$(-1)^{n_H} = -1$$

Parity conservation

Demonstration

$$d(t) = \sum_{n_H} d_{n_H} e^{-i n_H \omega t} \quad \omega T = 2\pi$$

$$\begin{aligned} d(t + \frac{T}{2}) &= \sum_{n_H} d_{n_H} e^{-i n_H \omega (t + \frac{T}{2})} \\ &= \sum_{n_H} d_{n_H} e^{-i n_H \omega t} e^{-i n_H \pi} \end{aligned}$$

$$d(t + \frac{T}{2}) = -d(t)$$

$$\sum_{n_H} d_{n_H} e^{-i n_H \omega t} e^{-i n_H \pi} = - \sum_{n_H} d_{n_H} e^{-i n_H \omega t}$$

$$d_{n_H} e^{-i n_H \pi} = -d_{n_H} \quad ; \quad d_{n_H} (1 + e^{-i n_H \pi}) = 0$$

$$e^{-i n_H \pi} = -1$$

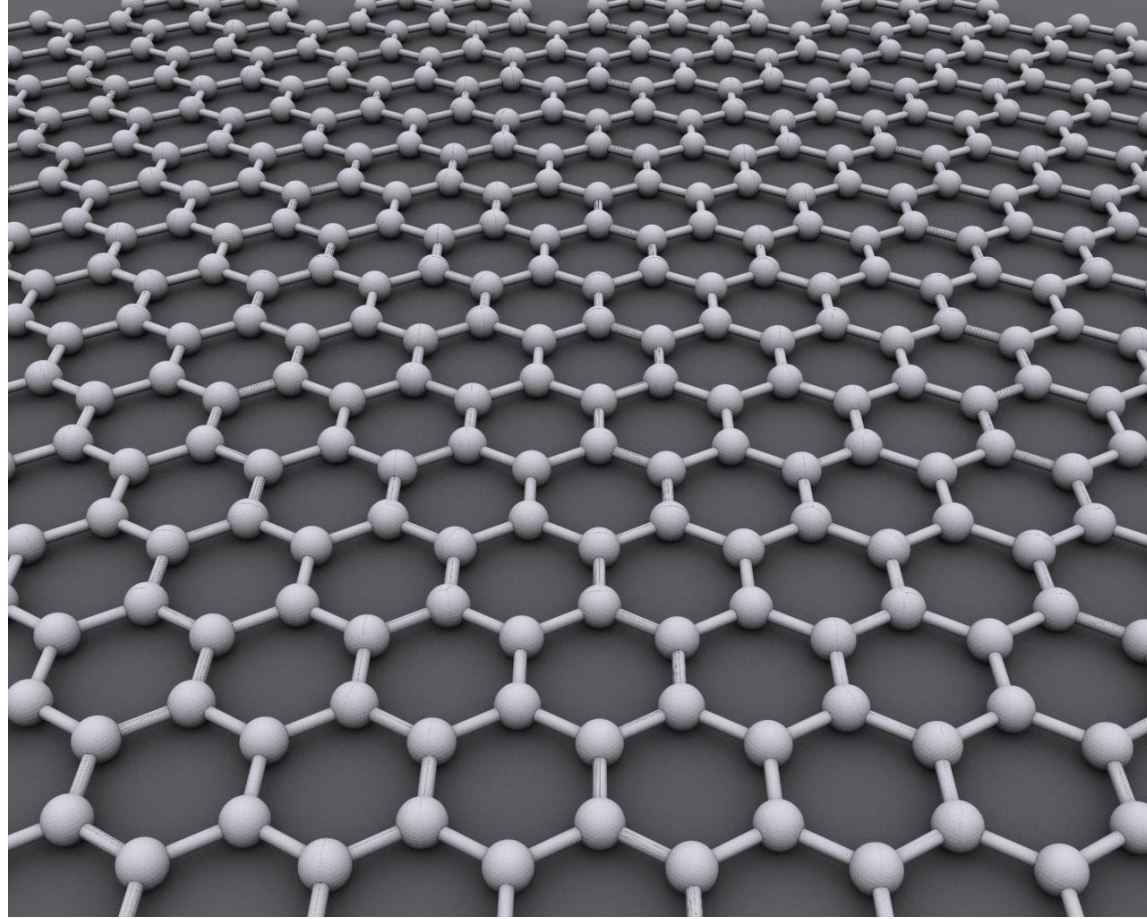
Using Euler's identity: $e^{-i n_H \pi} = \cos(n_H \pi) - i \sin(n_H \pi)$

$$\cos(n_H \pi) = -1$$

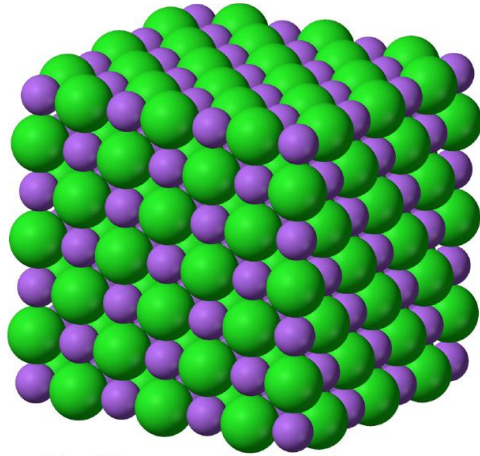
$$(-1)^{n_H} = 1 \quad \text{Parity conservation}$$

Anisotropy in laser-matter interactions

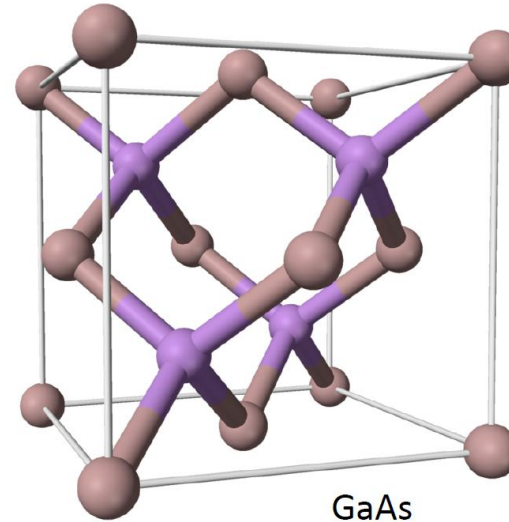
$x \leftrightarrow y \leftrightarrow z$



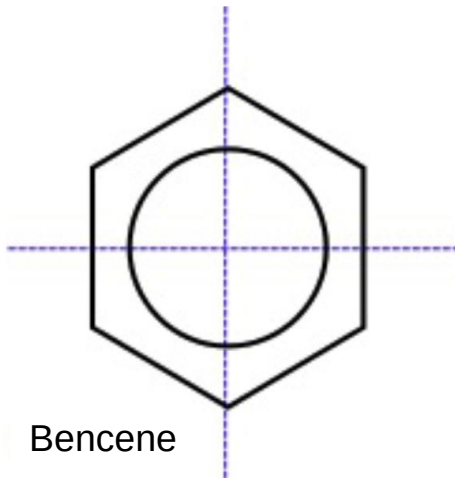
Inversion symmetry? Isotropy?



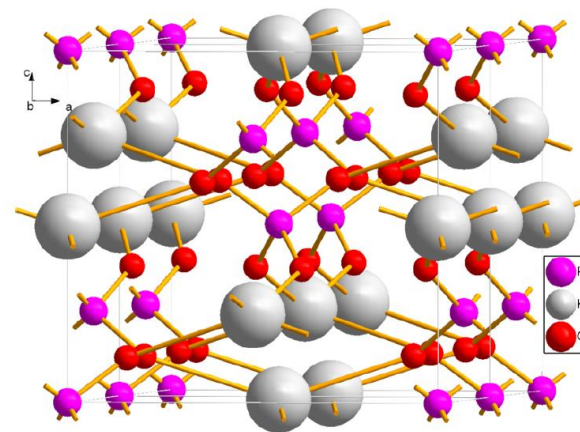
NaCl



GaAs



Bencene



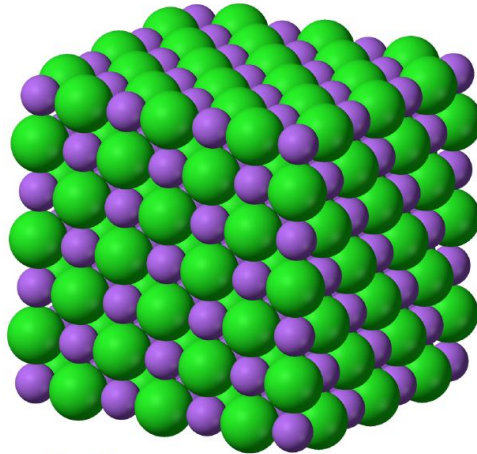
KDP

Inversion symmetry? Isotropy?

$$(x, y, z) \leftrightarrow (-x, -y, -z)$$

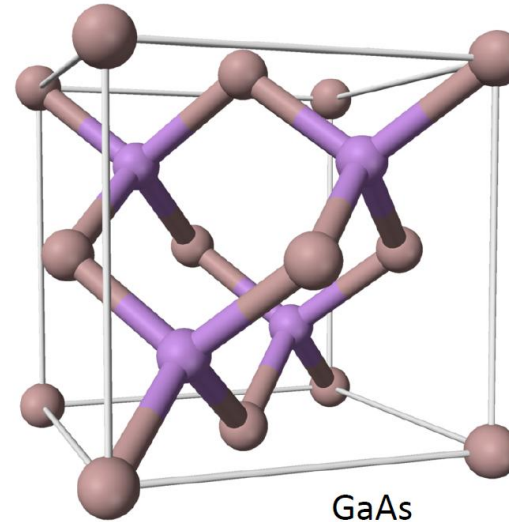
Centrosymmetric

Isotropic



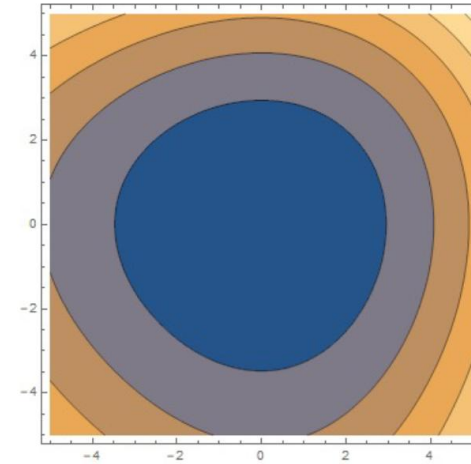
NaCl

No centrosymmetric

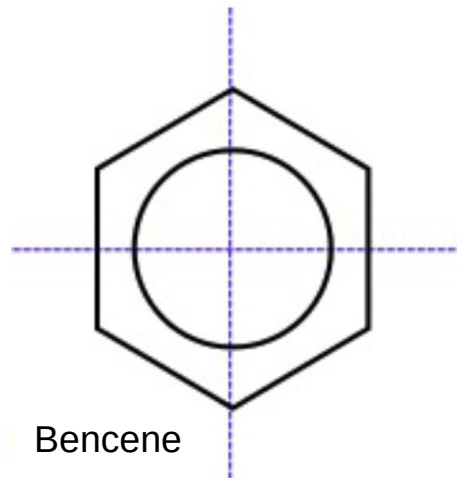


GaAs

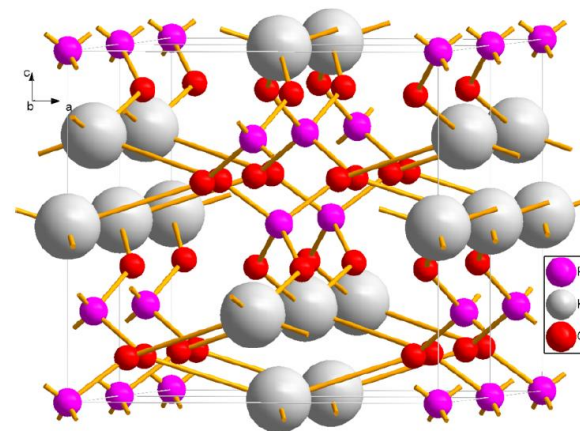
**Isotropic,
noncentrosymmetric
potential**



Anisotropic



Bencene

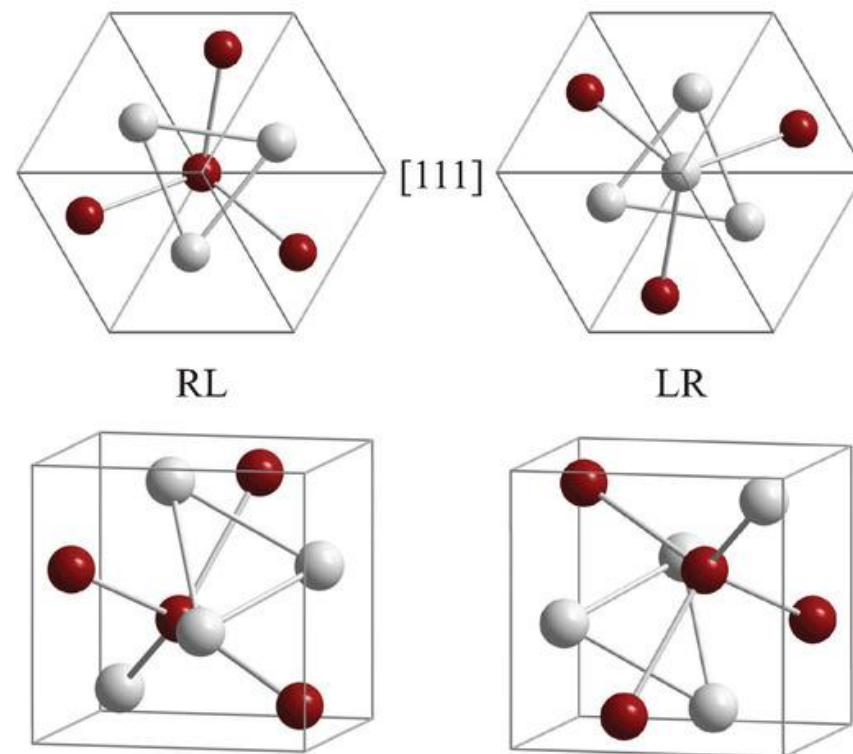
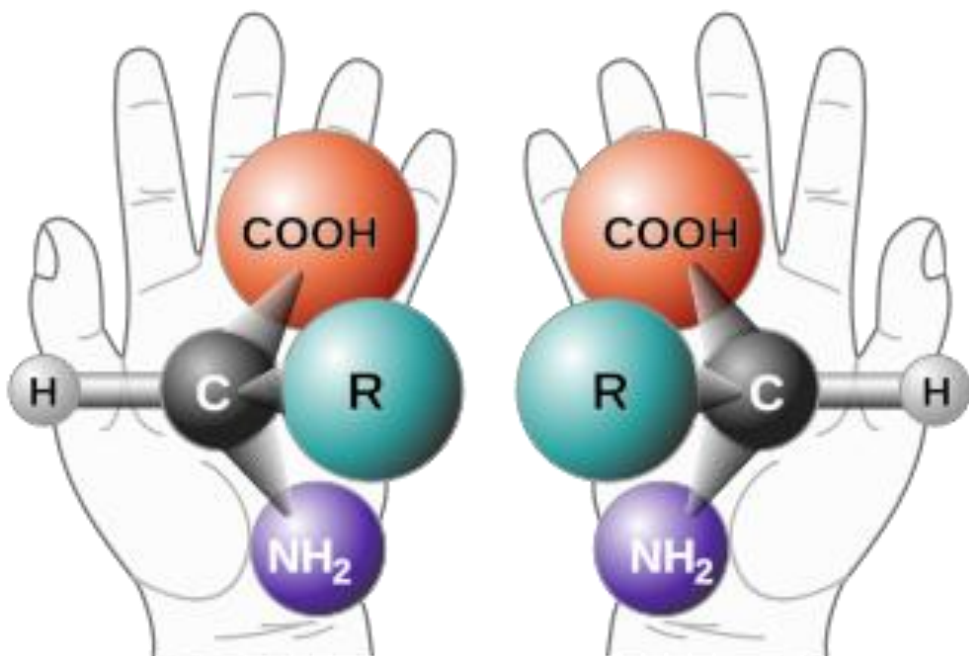


KDP

Chiral laser-matter interactions

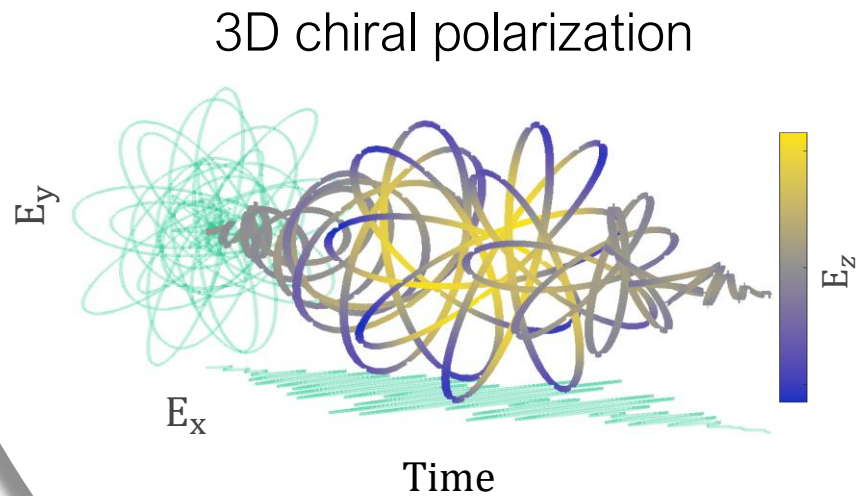
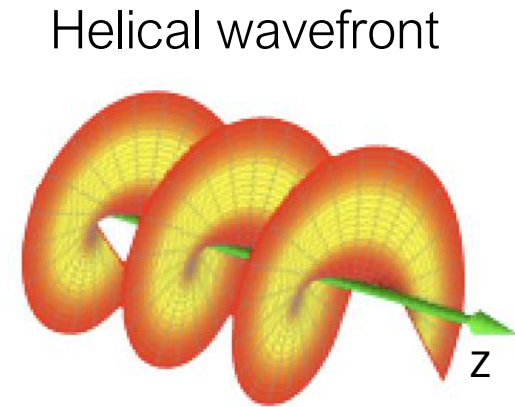
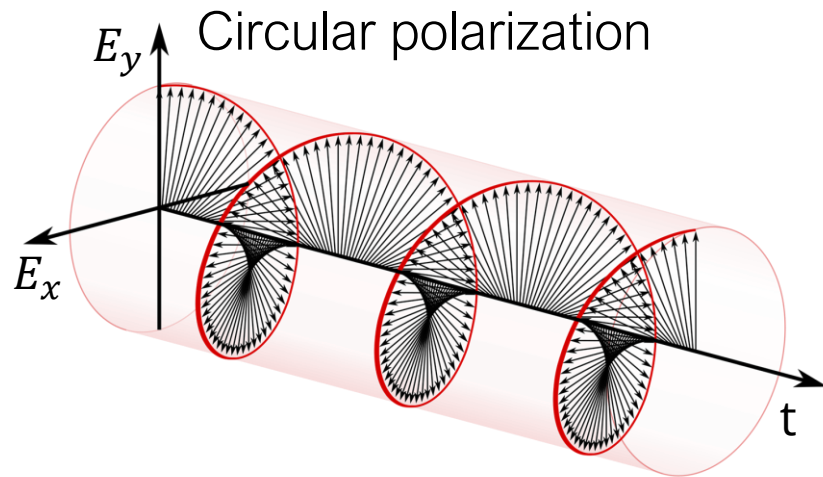
Chirality is the absence of mirror symmetry

Chiral matter

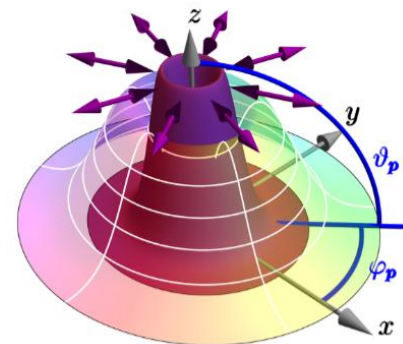


or/and chiral light

Overview of chiral light

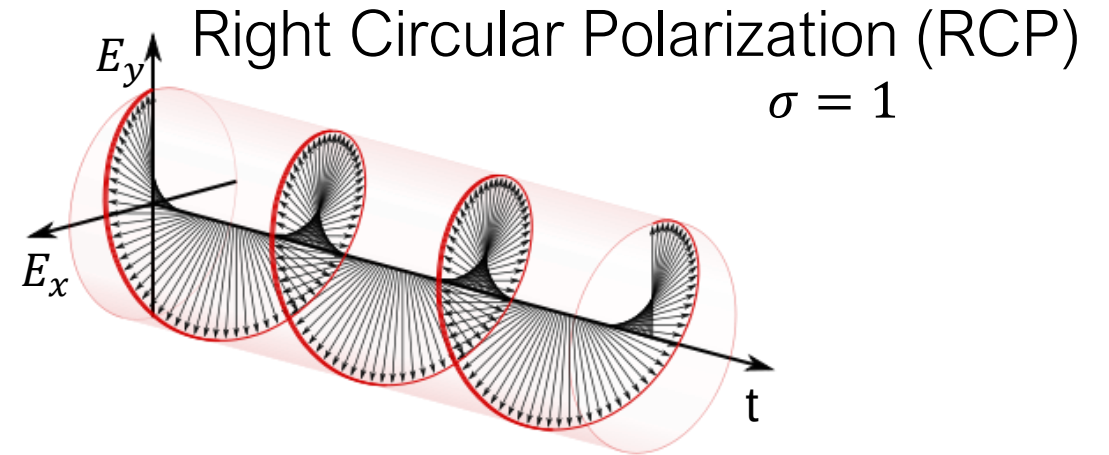
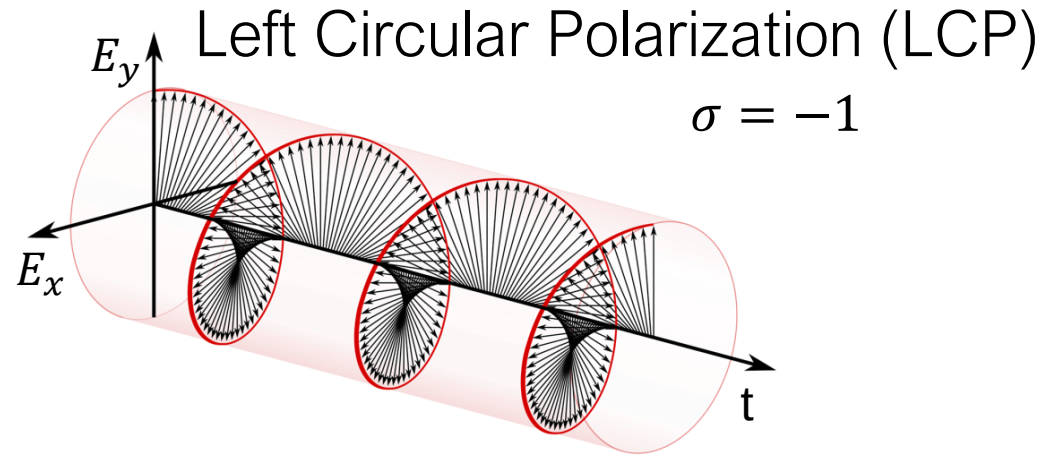


Spatial polarization pattern



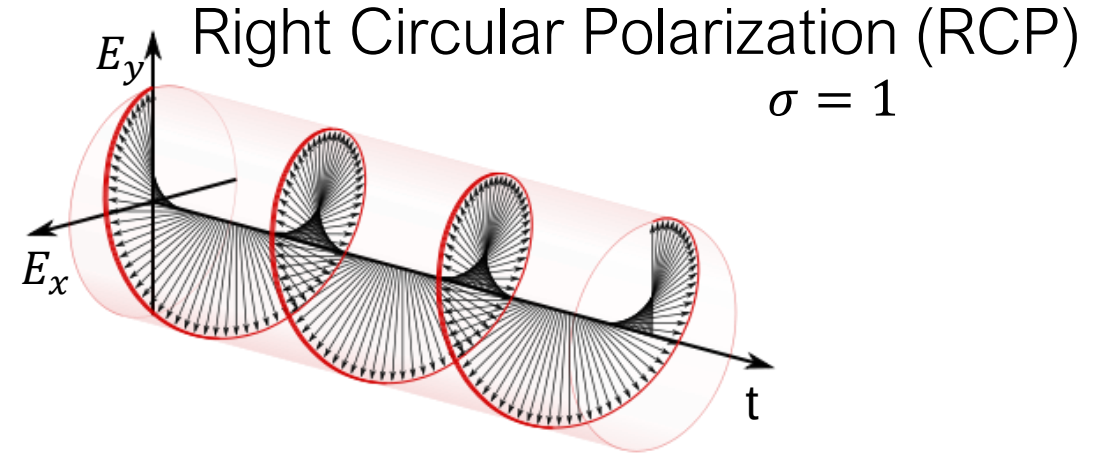
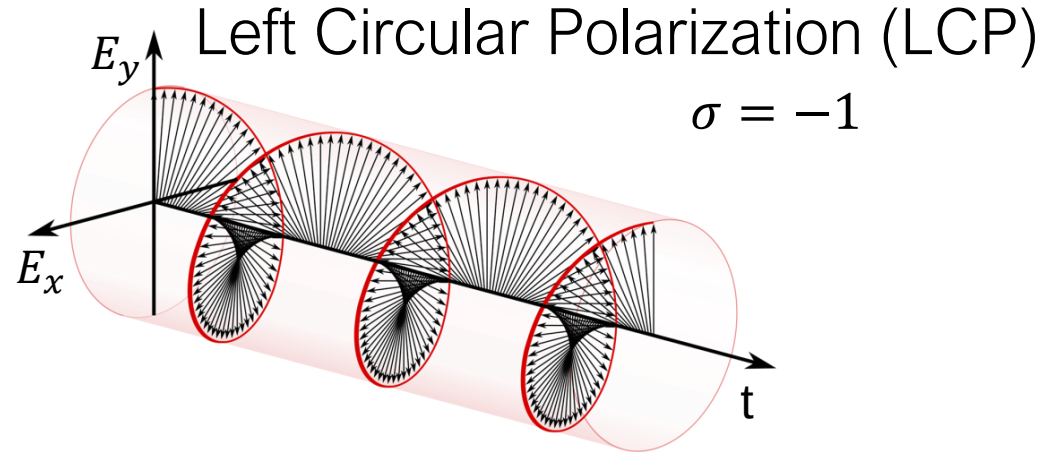
The angular momentum of light

Spin angular momentum (SAM) $\hbar\sigma$

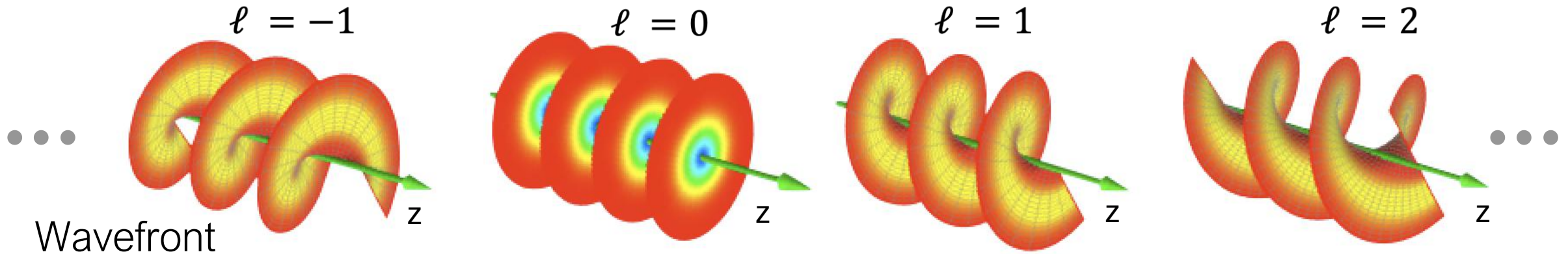


The angular momentum of light

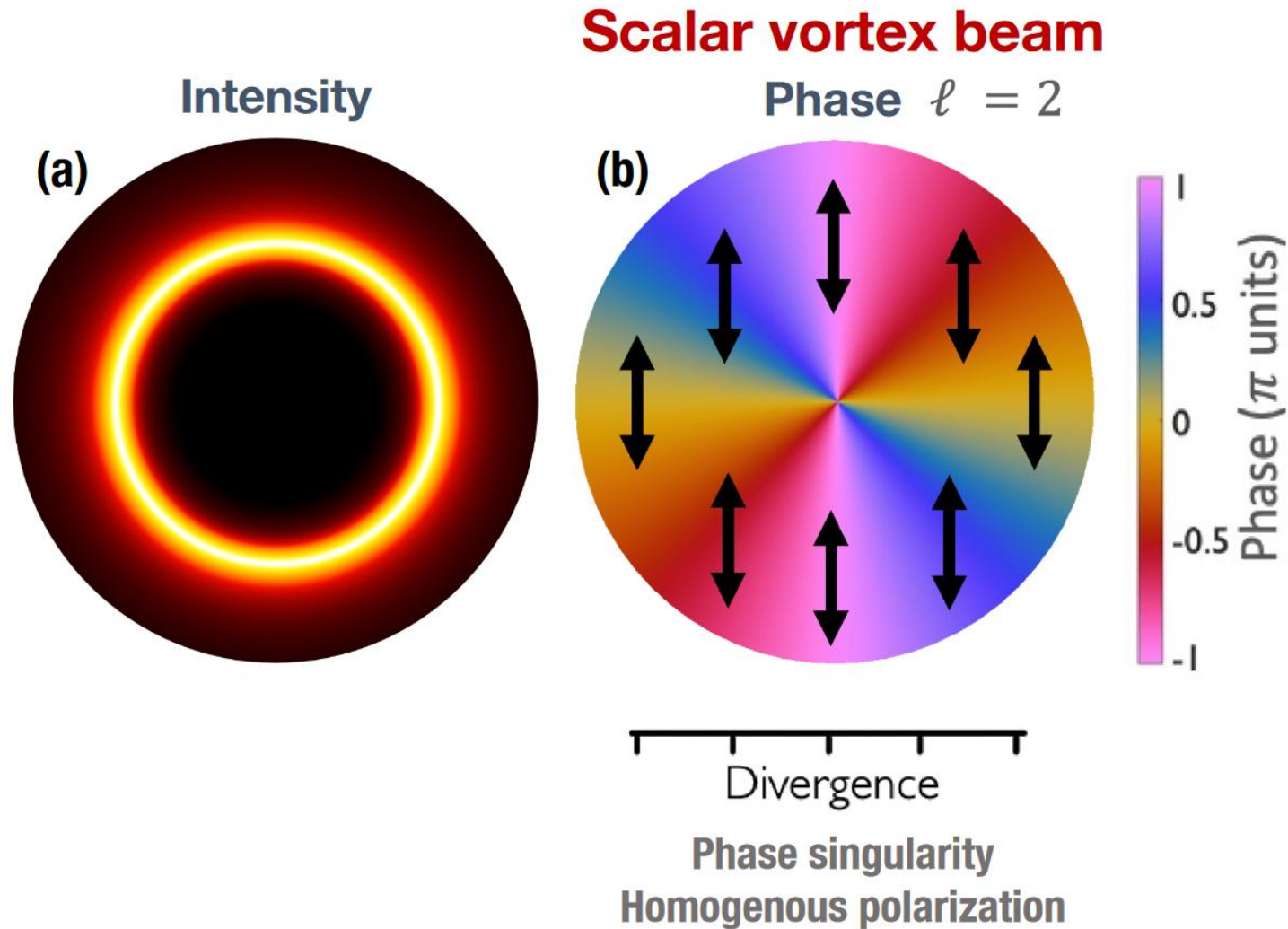
Spin angular momentum (SAM) $\hbar\sigma$



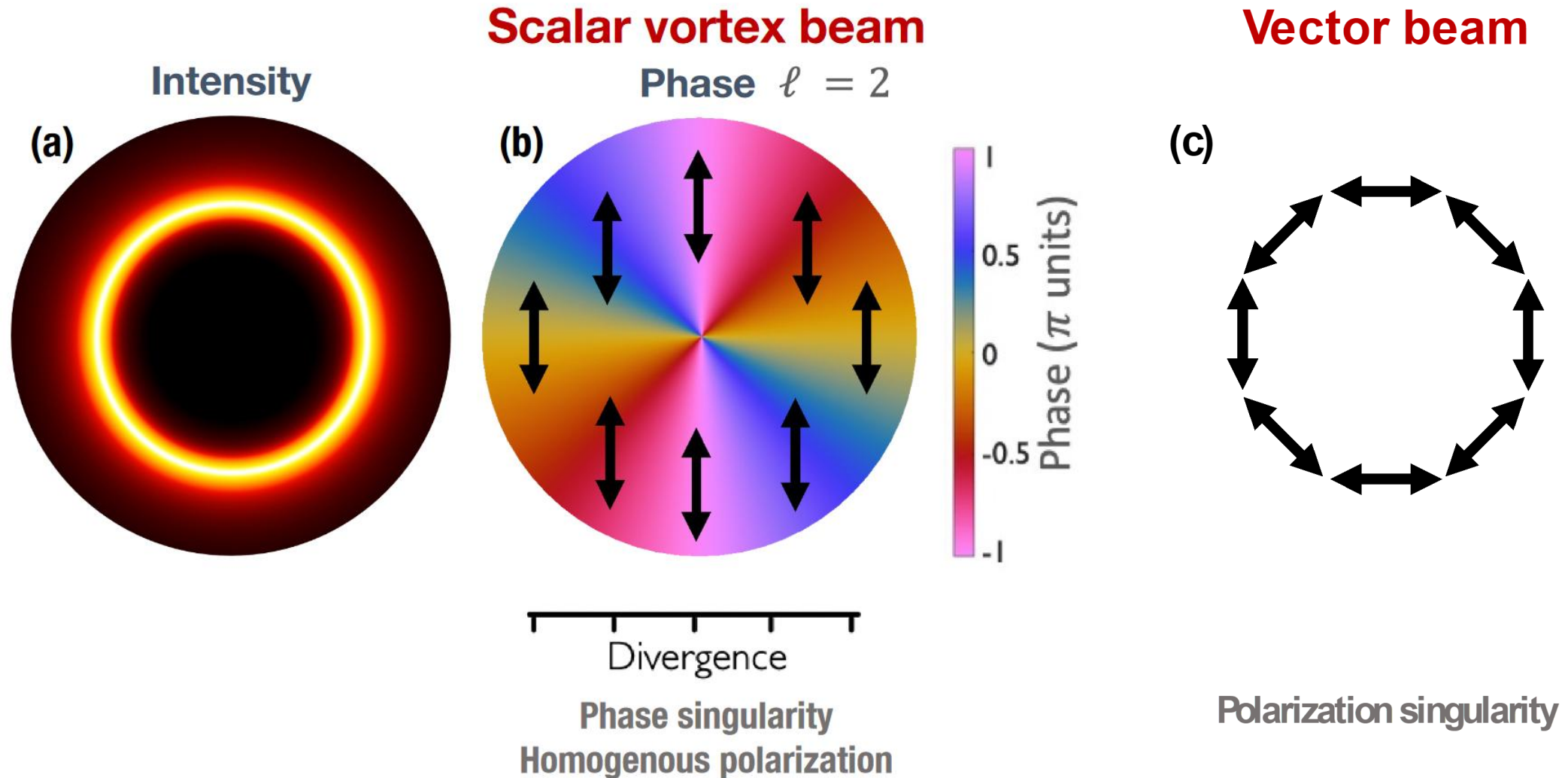
Orbital angular momentum (OAM) $\hbar\ell$



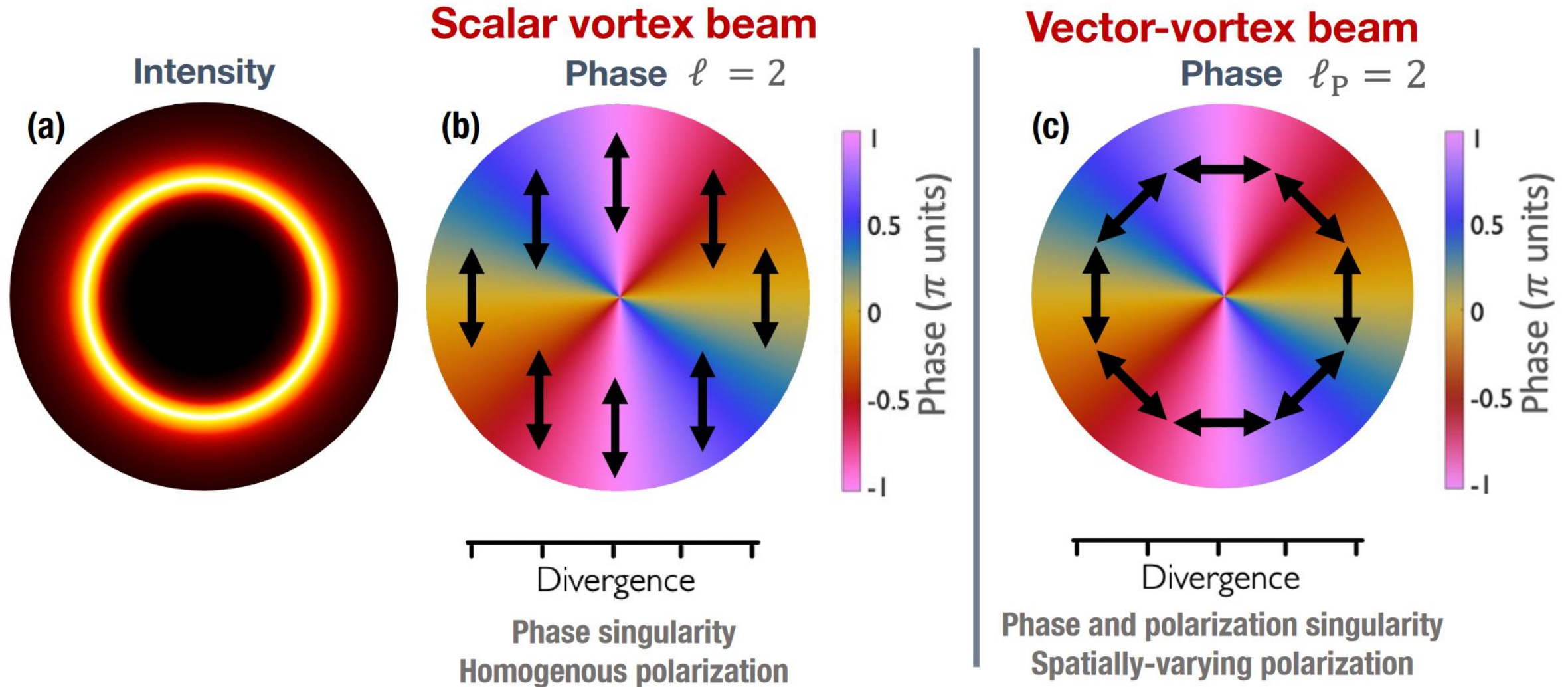
Scalar and vectorial vortex beams



Scalar and vectorial vortex beams



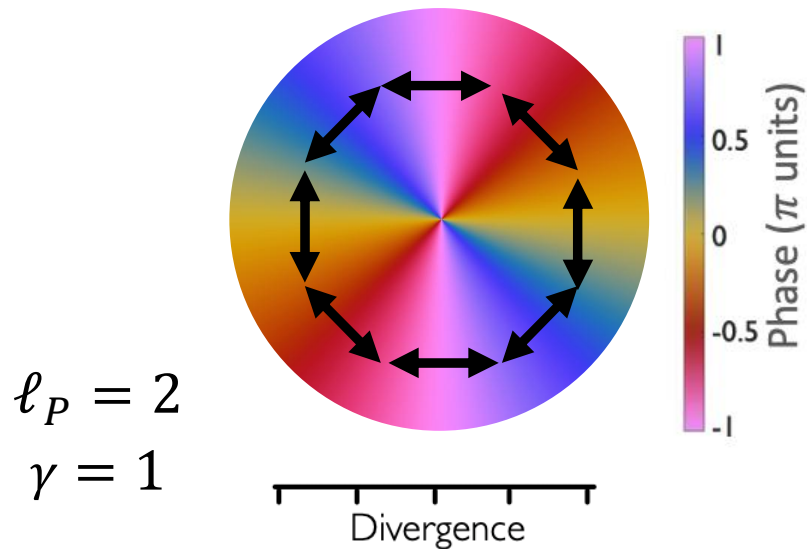
Scalar and vectorial vortex beams



Topological characterization of a **vector vortex beam**

Poincaré index and Pancharatnam topological charge of the driving vector-vortex beam

$$e^{-i\ell_P\phi} [\cos(\gamma\phi + \varphi) \mathbf{u}_x + \sin(\gamma\phi + \varphi) \mathbf{u}_y]$$



Pancharatnam-Berry phase

Pancharatnam topological charge

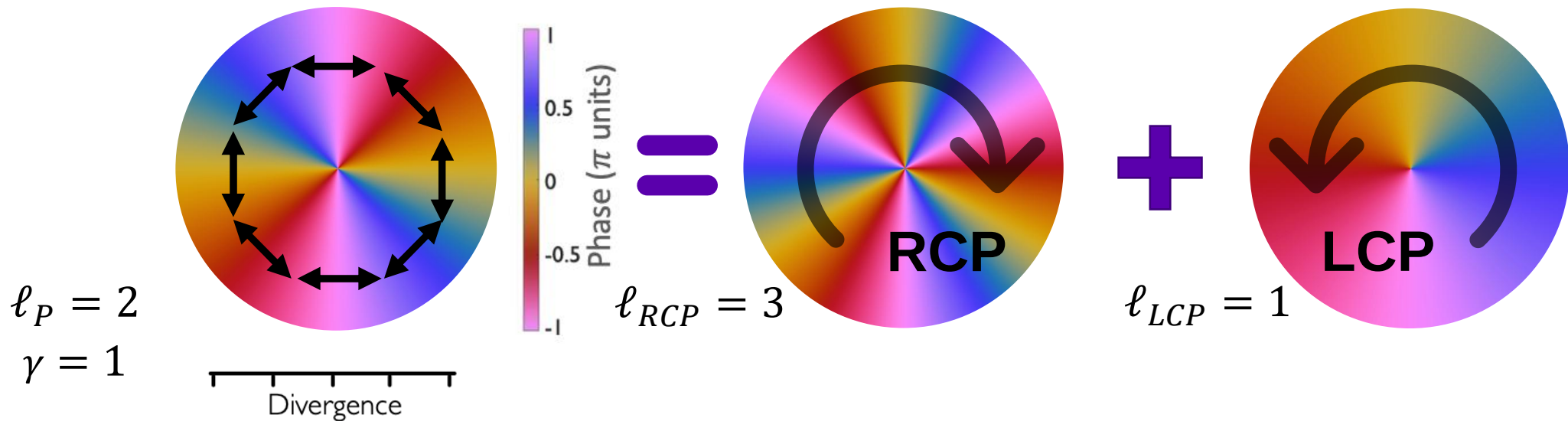
$$\varphi_P \equiv -\text{Arg}(\langle E(\phi = 0) | E \rangle) ; \quad \ell_P \equiv \frac{1}{2\pi} \oint_C d\varphi_P$$

A. Niv, et al., Opt. Express (2006)

Topological characterization of a **vector vortex beam**

Poincaré index and Pancharatnam topological charge of the driving vector-vortex beam

$$e^{-i\ell_P\phi} [\cos(\gamma\phi + \varphi) \mathbf{u}_x + \sin(\gamma\phi + \varphi) \mathbf{u}_y] \propto [e^{-i(\ell_P+\gamma)\phi} \overset{\curvearrowright}{\mathbf{u}}_{RCP} + e^{-i(\ell_P-\gamma)\phi} e^{i2\varphi} \overset{\curvearrowleft}{\mathbf{u}}_{LCP}]$$



Pancharatnam-Berry phase

$$\varphi_P \equiv -\text{Arg}(\langle E(\phi = 0) | E \rangle) ;$$

Pancharatnam topological charge

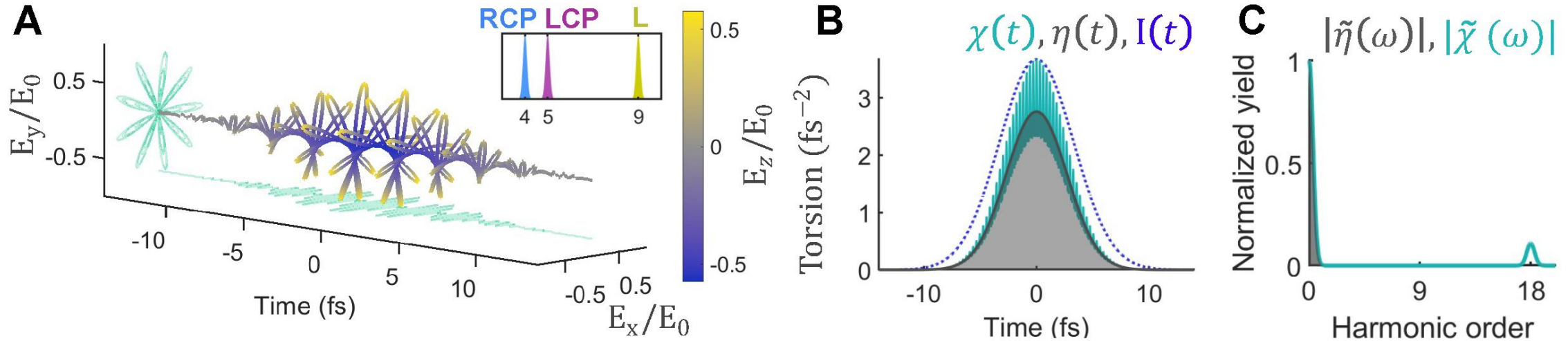
$$\ell_P \equiv \frac{1}{2\pi} \oint_C d\varphi_P = \frac{\ell_{RCP} + \ell_{LCP}}{2}$$

Locally-chiral light

3D chiral polarization

The electric field exhibits instantaneous torsion:

$$\chi(t) = \frac{\vec{E}(t)}{E_0^3} \cdot \left(\frac{d\vec{E}(t)}{dt} \times \frac{d^2\vec{E}(t)}{dt^2} \right)$$



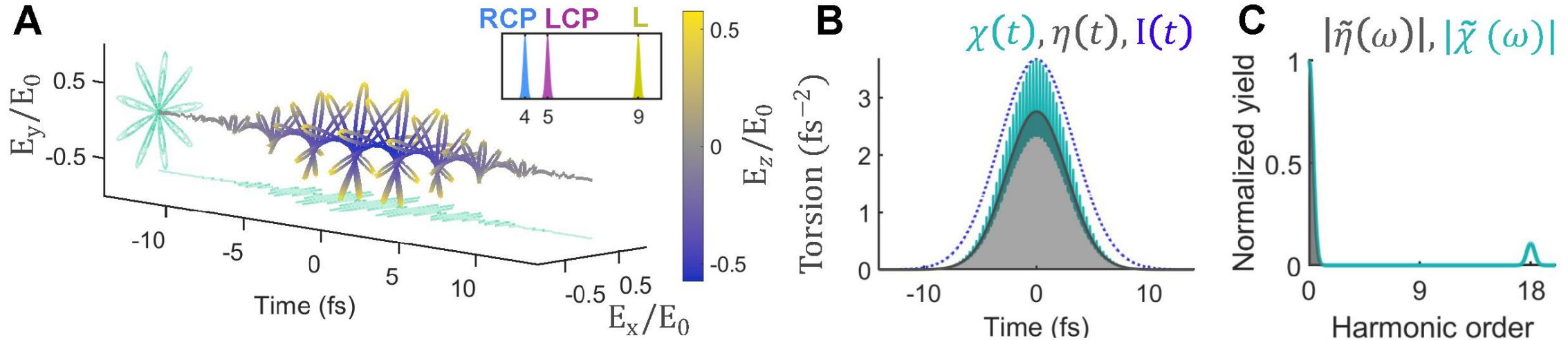
A. de las Heras *et al.* *Nano Letters* (2026)

Locally-chiral light

3D chiral polarization

The electric field exhibits instantaneous torsion:

$$\chi(t) = \frac{\vec{E}(t)}{E_0^3} \cdot \left(\frac{d\vec{E}(t)}{dt} \times \frac{d^2\vec{E}(t)}{dt^2} \right)$$



- The electric field vector is chiral at every spatial point
- Efficient chiral dipolar interaction, independently of the spatial scale
- The propagation of the z-polarization component is restricted in free space (transversality condition for plane waves $\vec{k} \cdot \vec{E} = 0$)
- Requires a multi-color synthesis
- Not measured in experiments yet

D. Ayuso *et al.* *Nat. Photonics* (2019)
A. de las Heras *et al.* *Nano Letters* (2026)



**Are there chiral light sources
in the high-frequency regime?**

Chiral light beams from free-electron lasers

SAM control

RESEARCH ARTICLE | MARCH 22 2016

Femtosecond X-ray magnetic circular dichroism absorption spectroscopy at an X-ray free electron laser

Daniel J. Higley; Konstantin Hirsch; Georgi L. Dakovski; Emmanuelle Jal; Edwin Yuan; Tianmin Liu; Alberto A. Lutman; James P. MacArthur; Elke Arenholz; Zhao Chen; Giacomo Coslovich; Peter Denes; Patrick W. Granitzka; Philip Hart; Matthias C. Hoffmann; John Joseph; Loïc Le Guyader; Ankush Mitra; Stefan Moeller; Hendrik Ohldag; Matthew Seaberg; Padraic Shafer; Joachim Stöhr; Arata Tsukamoto; Heinz-Dieter Nuhn; Alex H. Reid; Hermann A. Dürr; William F. Schlotter

RESEARCH ARTICLE | AUGUST 31 2016

Circular dichroism measurements at an x-ray free-electron laser with polarization control

G. Hartmann; A. O. Lindahl; A. Knie; N. Hartmann; A. A. Lutman; J. P. MacArthur; I. Shevchuk; J. Buck; A. Galler; J. M. Glowina; W. Helml; Z. Huang; N. M. Kabachnik; A. K. Kazansky; J. Liu; A. Marinelli; T. Mazza; H.-D. Nuhn; P. Walter; J. Viehhaus; M. Meyer; S. Moeller; R. N. Coffee; M. Ilchen

Structural Dynamics

ARTICLE

pubs.aip.org/aip/sdy

First experiments with ultrashort, circularly polarized soft x-ray pulses at FLASH2

Cite as: Struct. Dyn. 12, 034301 (2025); doi:10.1063/4.0000298

Submitted: 31 January 2025 · Accepted: 24 April 2025 ·

Published Online: 8 May 2025



S. Marotzke,^{1,2} D. Gupta,³ R.-P. Wang,¹ M. Pavelka,⁴ S. Dziarzhyski,¹ C. von Korff Schmising,⁵ S. Jana,⁵ N. Thielemann-Kühn,⁶ T. Amrhein,⁶ M. Weinelt,⁶ I. Vaskivskyi,⁷ R. Knut,⁴ D. Engel,⁵ M. Braune,¹ M. Ilchen,^{1,8,9} S. Savio,^{8,10} T. Otto,^{1,8} K. Tiedtke,¹ V. Scheppe,^{3,6} J. Rönsch-Schulenberg,¹ E. Schneidmiller,¹ C. Schüßler-Langeheine,³ H. A. Dürr,⁴ M. Beye,^{1,11} C. Brenner,¹ and N. Pontius^{2,a)}

OAM control (in the EUV range for experiments)

PHYSICAL REVIEW X 7, 031036 (2017)

Extreme-Ultraviolet Vortices from a Free-Electron Laser

Primož Rebernik Ribič,^{1,*} Benedikt Rösner,² David Gauthier,¹ Enrico Allaria,¹ Florian Döring,² Laura Foglia,¹ Luca Giannessi,^{1,4} Nicola Mahne,¹ Michele Manfreda,¹ Claudio Masciovecchio,¹ Riccardo Mincigrucci,¹ Najmeh Mirian,¹ Emiliano Principi,¹ Eléonore Roussel,¹ Alberto Simoncig,¹ Simone Spampinati,¹ Christian David,² and Giovanni De Nino^{1,3}

¹*Elettra-Sincrotrone Trieste, Strada Statale 14-km 163,5, 34149 Basovizza, Trieste, Italy*

²*Paul Scherrer Institut, 5232 Villigen-PSI, Switzerland*

³*Laboratory of Quantum Optics, University of Nova Gorica, 5001 Nova Gorica, Slovenia*

⁴*ENEA, C.R. Frascati, Via E. Fermi 45, 00044 Frascati (Rome), Italy*

(Received 20 June 2017; published 28 August 2017; corrected 19 July 2018)

Chiral light beams from free-electron lasers

Theory proposals for OAM control in the X-ray regime



Attosecond X-ray pulse pairs carrying distinct topological charges

Chenzhi Xu,^{1,2,3} Jiawei Yan,^{4,*} Gianluca Geloni,^{3,†} Christoph Lechner,³ and Haixiao Deng^{5,‡}

¹Shanghai Institute of Applied Physics, Chinese Academy of Sciences, Shanghai 201800, China

²University of Chinese Academy of Sciences, Beijing 100049, China

³European XFEL, 22869 Schenefeld, Germany

⁴Deutsches Elektronen-Synchrotron DESY, 22603 Hamburg, Germany

⁵Shanghai Advanced Research Institute, Chinese Academy of Sciences, Shanghai 201210, China

[arXiv:2508.19020](https://arxiv.org/abs/2508.19020)

Chiral light beams from Plasma-Based Soft-X-Ray Lasers

SAM control

PRL 115, 083901 (2015)

PHYSICAL REVIEW LETTERS

week ending
21 AUGUST 2015



Demonstration of a Circularly Polarized Plasma-Based Soft-X-Ray Laser

A. Depresseux,¹ E. Oliva,² J. Gautier,¹ F. Tissandier,¹ G. Lambert,¹ B. Vodungbo,¹ J-P. Goddet,¹ A. Tafzi,¹ J. Nejdil,³

M. Kozlova,³ G. Maynard,² H. T. Kim,^{4,5} K. Ta Phuoc,¹ A. Rousse,¹ P. Zeitoun,¹ and S. Sebban¹

¹Laboratoire d'Optique Appliquée, ENSTA ParisTech, CNRS, École Polytechnique, Université Paris-Saclay,
828 boulevard des Maréchaux, 91762 Palaiseau, France

²Laboratoire de Physique des Gaz et des Plasmas, CNRS Université Paris-Sud 11, 91405, Orsay, France

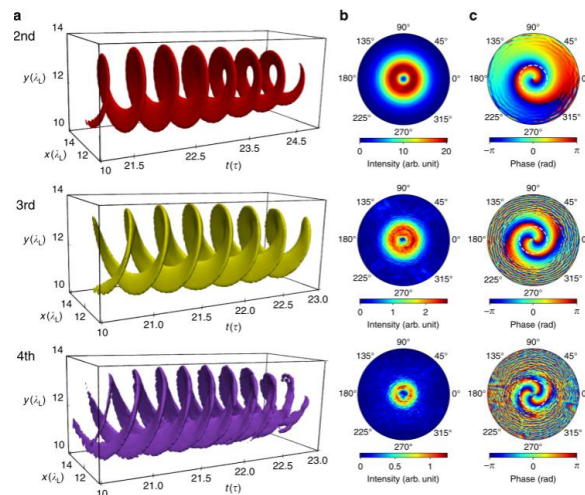
³ELI Beamlines Project, Institute of Physics of the ASCR, Na Slovance 2, 182 21 Prague 8, Czech Republic

⁴Advanced Photonics Research Institute, GIST, Gwangju 500-712, Korea

⁵Center for Relativistic Laser Science, Institute for Basic Science (IBS), Gwangju 500-712, Korea

(Received 27 May 2015; published 19 August 2015)

OAM control in harmonic beams via relativistic oscillating mirror (theory)

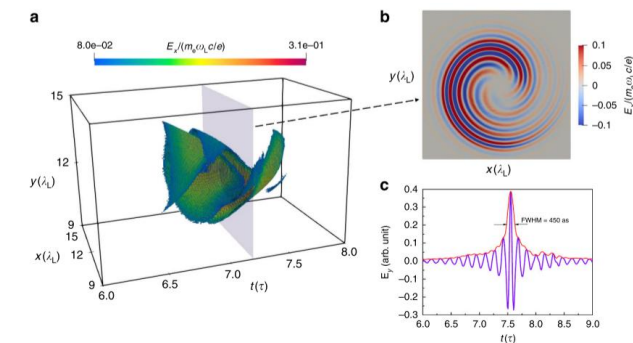


Article | [Open access](#) | Published: 05 December 2019

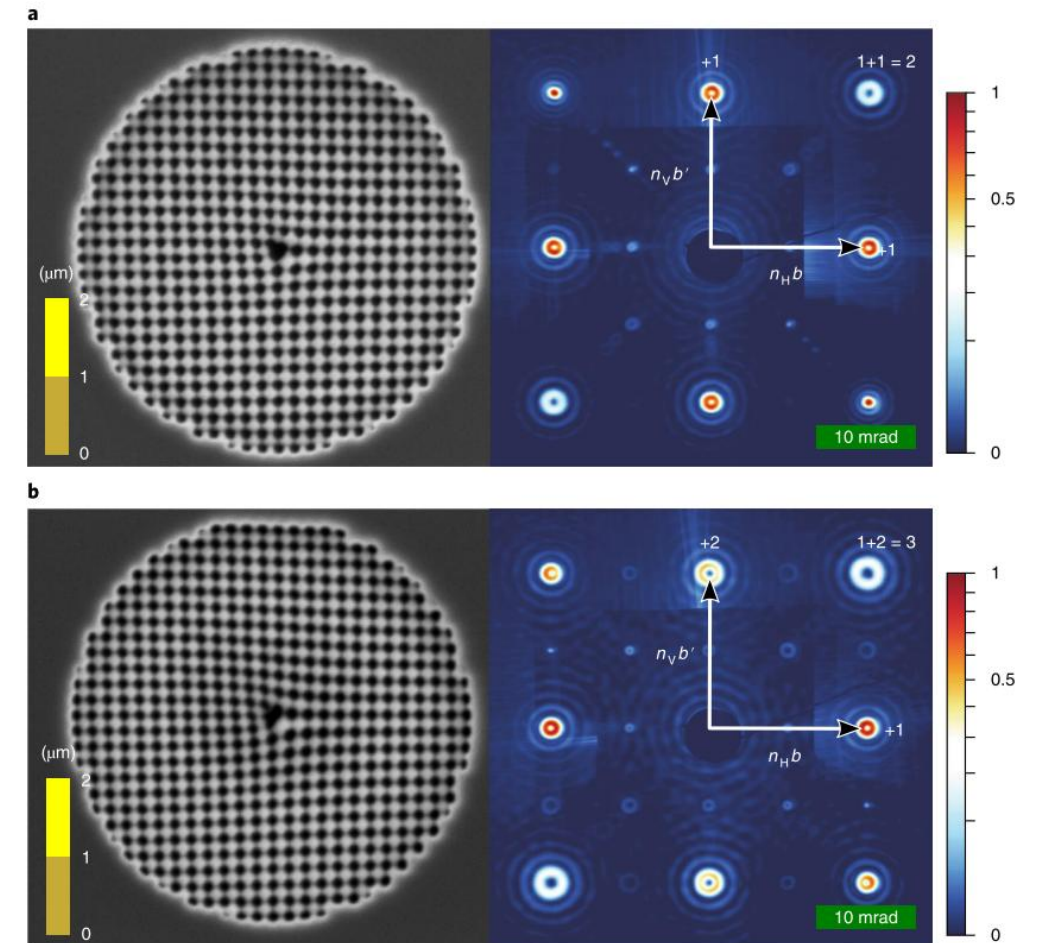
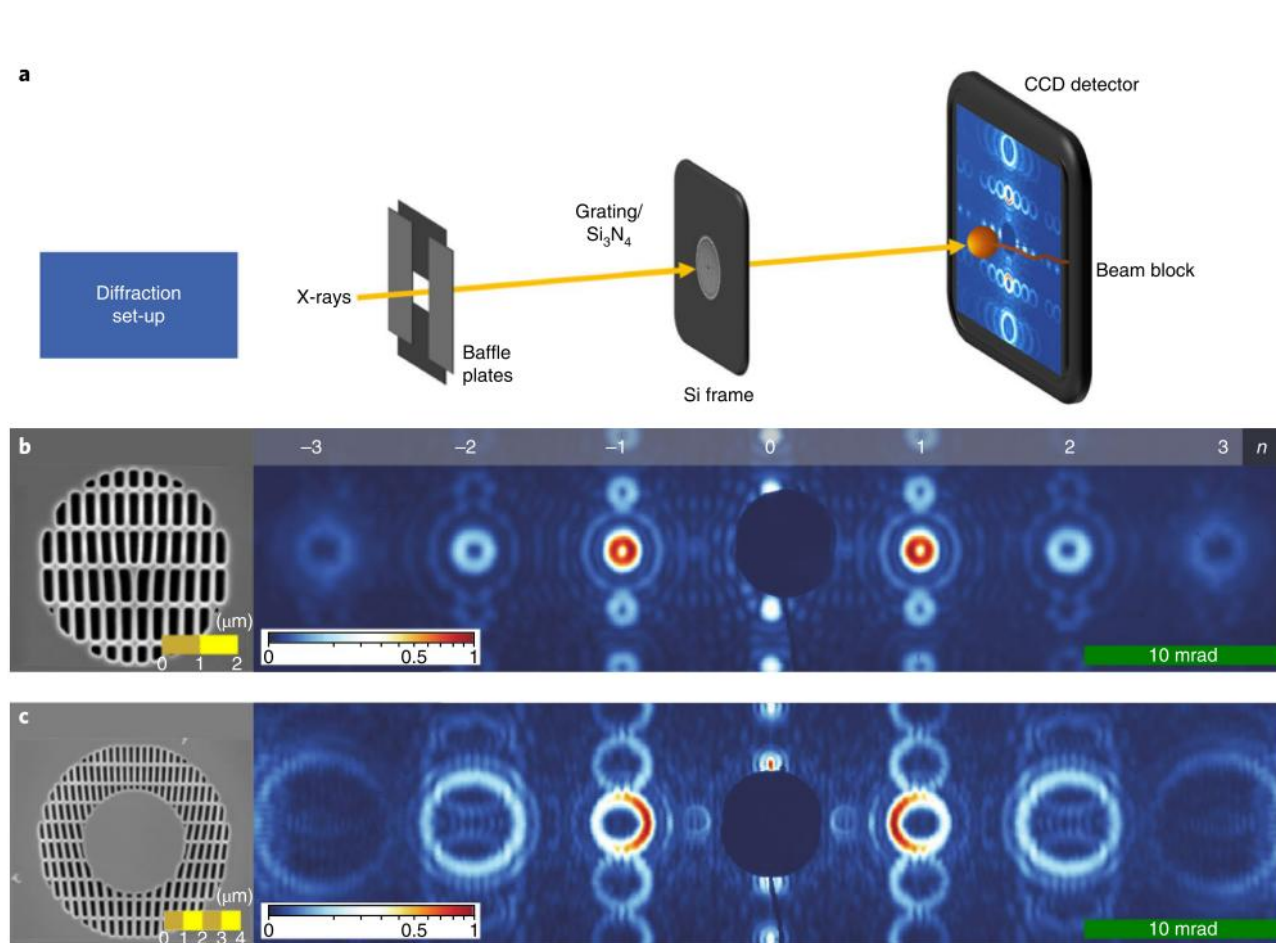
Intense attosecond pulses carrying orbital angular momentum using laser plasma interactions

[J. W. Wang](#) , [M. Zepf](#) & [S. G. Rykovanov](#)

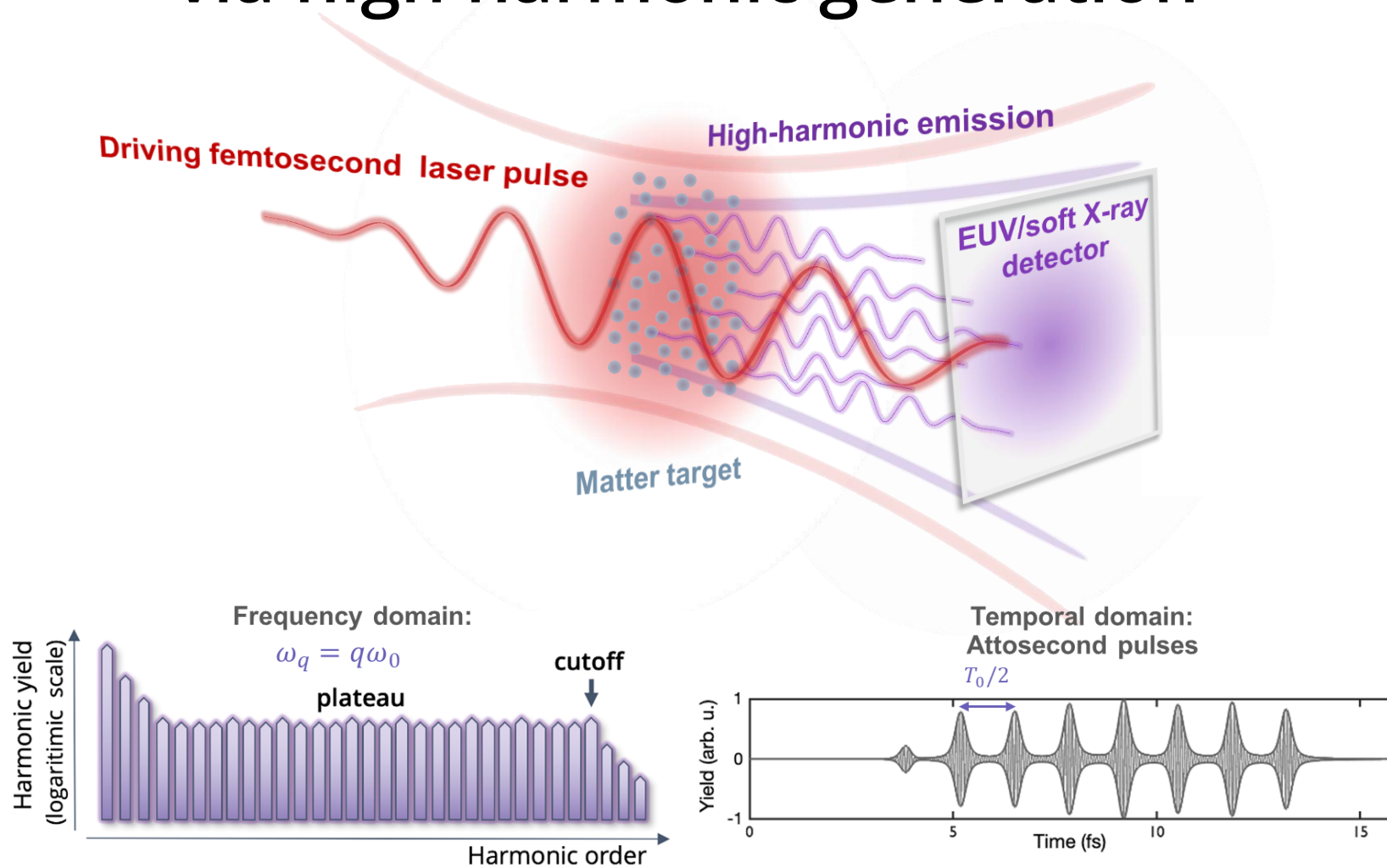
[Nature Communications](#) 10, Article number: 5554 (2019) | [Cite this article](#)



OAM control via soft X-ray diffractive optics



Tailoring EUV light via high harmonic generation



Dipole selection rules in high harmonic generation

Conservation laws applied to the standard one-driver HHG in atoms:

Photon energy conservation: $\hbar\omega_q = q \hbar\omega_0$ q is the harmonic order

Linear momentum conservation: $\vec{k}_q = q \vec{k}$

Orbital angular momentum conservation: $\ell_q = q \ell$

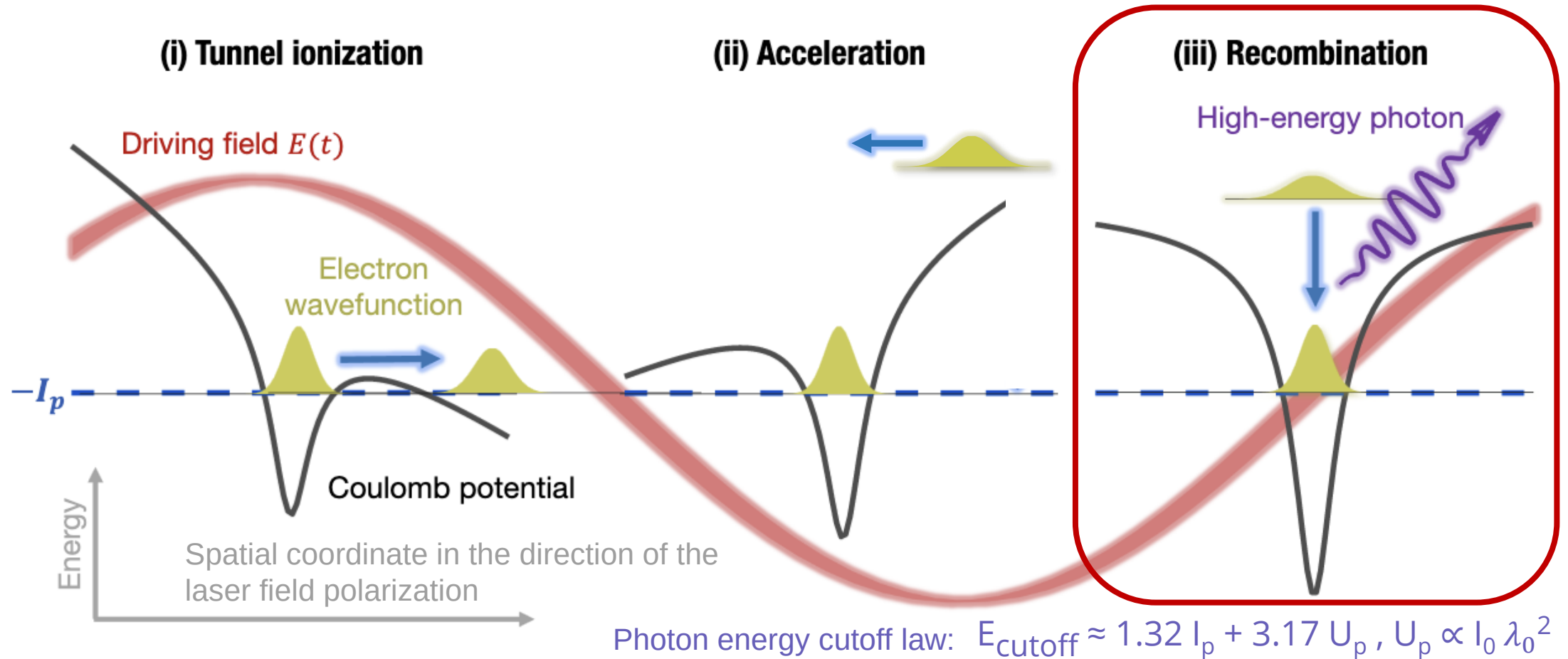
Spin angular momentum conservation: $\sigma_q = q \sigma$ Spin states can only be: -1, 1 or 0

Harmonics cannot be generated from a driver with $\sigma = -1$ or 1 !

High-order harmonic generation (HHG) at the atomic level

Three-step process

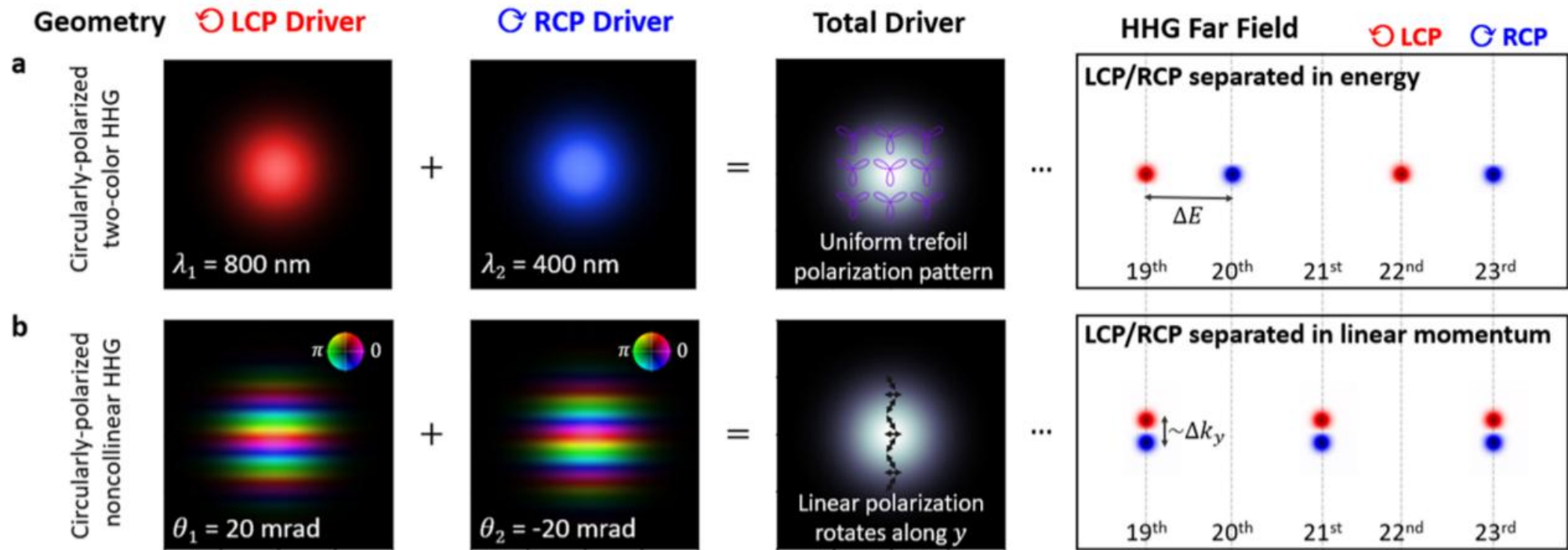
Only in the case of a low-ellipticity driver!



K. J. Schafer et al. *Phys. Rev. Lett.* (1993)
P. B. Corkum. *Phys. Rev. Lett.* (1993); M. Lewenstein et al. *Phys. Rev. A* (1994)

J. Krause et al. *Phys. Rev. Lett.* (1992)
M. Lewenstein et al. *Phys. Rev. A* (1994)

Generation of circularly polarized high harmonic beams



A. Fleischer, et al.
Nat. Photonics (2014)
O. Kfir, et al.
Nat. Photonics (2015)

D. Hickstein et al.
Nat. Photonics (2015)

Circularly polarized two-color HHG

$$\omega_q = q \omega_1$$

$$\omega_q = n_1 \omega_1 + n_2 \omega_2 = (n_1 + 2n_2) \omega_1 \rightarrow q = n_1 + 2n_2$$

$\omega_2 = 2\omega_1$

$$\sigma_q = n_1 \sigma_1 + n_2 \sigma_2 = (n_1 - n_2) \sigma_1$$

$\sigma_2 = -\sigma_1$

Two possible channels:

$$\sigma_q = \sigma_1, \quad n_2 = n_1 - 1, \quad q = (3n_1 - 2)$$

$$\sigma_q = -\sigma_1, \quad n_2 = n_1 + 1, \quad q = (3n_1 + 2)$$

$$\text{Example: } n_1 = 7, n_2 = 6 \rightarrow \sigma_q = \sigma_1, q = 19$$

$$n_1 = 6, n_2 = 7 \rightarrow \sigma_q = -\sigma_1, q = 20$$

Circularly polarized noncollinear HHG

$$q = n_1 + n_2$$

$$\sigma_q = n_1 \sigma_1 + n_2 \sigma_2 \stackrel{q}{=} (n_1 - n_2) \sigma_1$$

$$\sigma_2 = -\sigma_1$$

Two possible channels:

$$\sigma_q = \sigma_1, \quad n_2 = n_1 - 1, \quad q = 2n_1 - 1$$

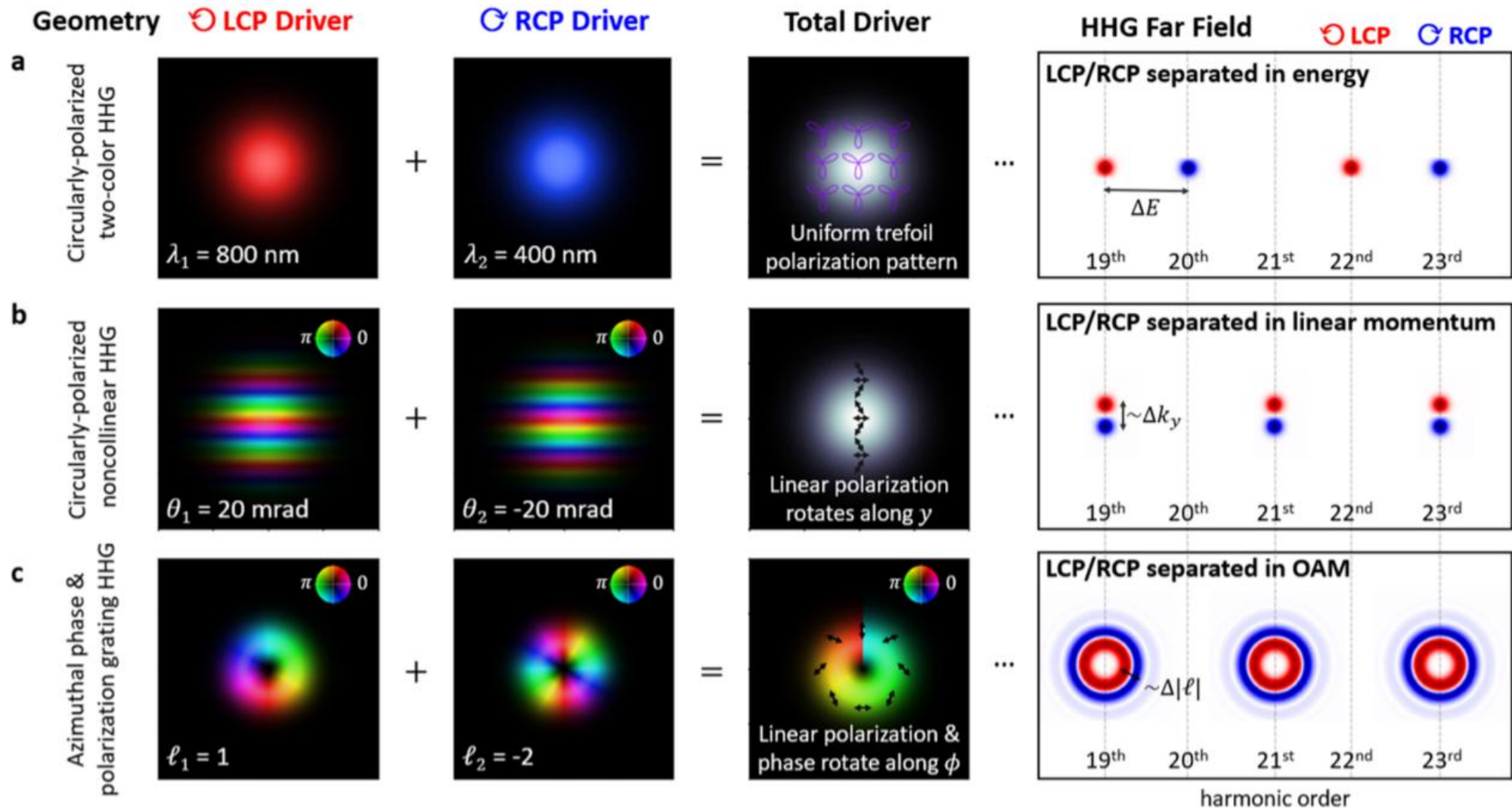
$$\sigma_q = -\sigma_1, \quad n_2 = n_1 + 1, \quad q = 2n_1 + 1$$

$$\text{Example: } n_1 = 10, \quad n_2 = 9 \rightarrow \sigma_q = \sigma_1, \quad q = 19$$

$$n_1 = 9, \quad n_2 = 10 \rightarrow \sigma_q = -\sigma_1, \quad q = 19$$

$$\vec{k}_q = n_1 \vec{k}_1 + n_2 \vec{k}_2, \quad \vec{k}_1 \neq \vec{k}_2$$

Generation of circularly polarized high harmonic beams



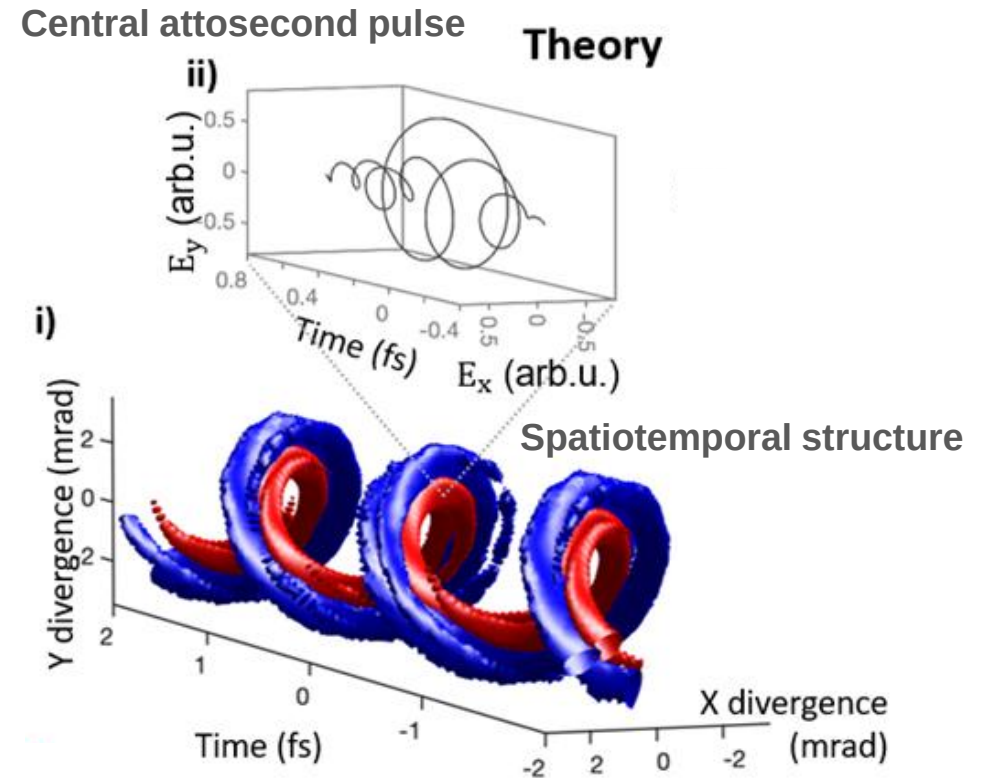
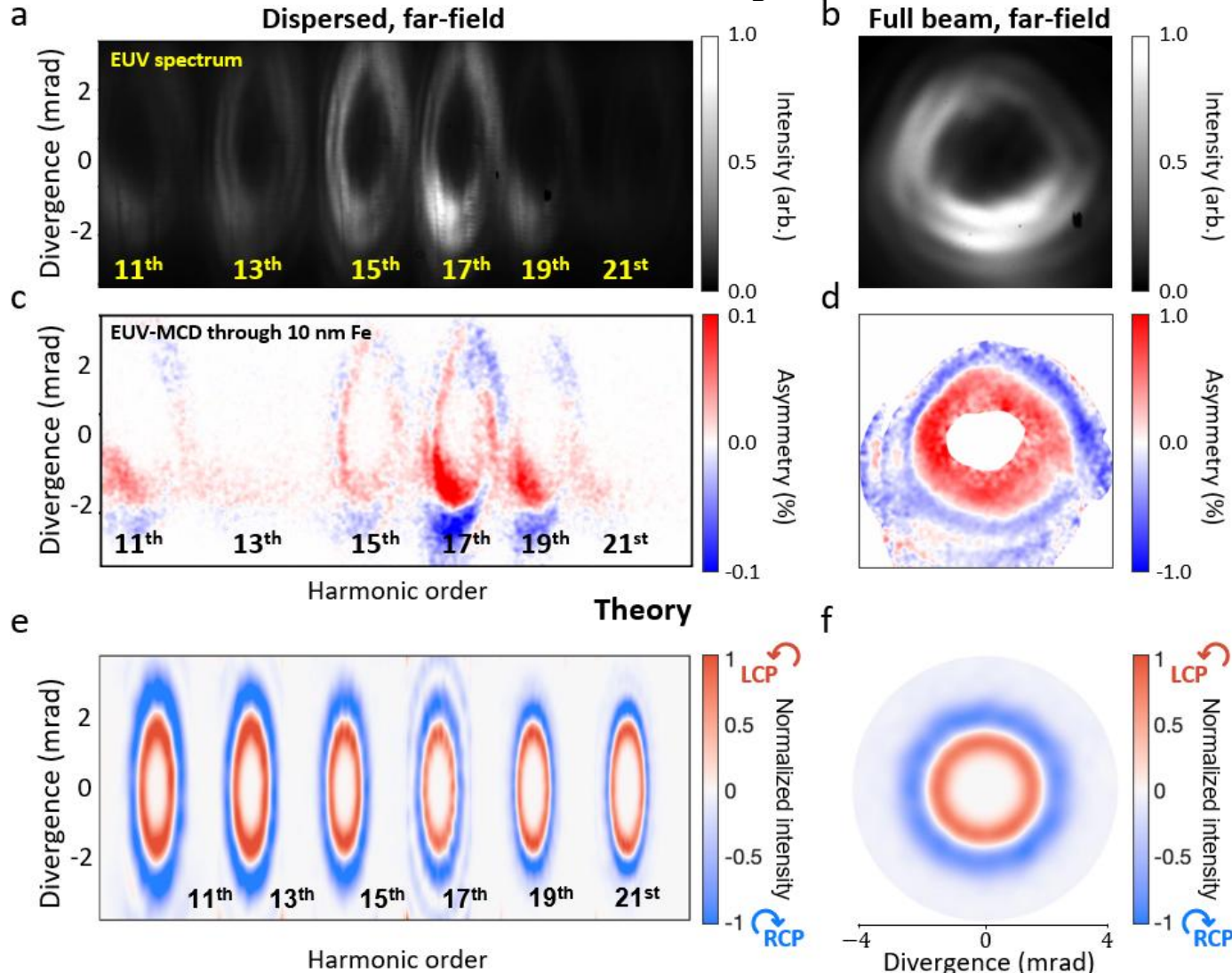
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Nat. Photonics (2015)

D. Hickstein et al.
Nat. Photonics (2015)

N. Brooks et al.
ACS Photonics (2025)

Generation of circularly polarized high harmonic beams and attosecond pulse trains from a vector-vortex driver

Experiment **JILA**



N. Brooks *et al.* *ACS Photonics* (2025)

High harmonic structured beams in the extreme ultraviolet

Pioneer HHG experiment on OAM

M. Zürch *et al. Nat. Physics* (2012)

OAM selection rule in HHG

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A. K. Pandey *et al., ACS Photonics* (2022)

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A. de las Heras *et al. Optica* (2024)

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D. Hickstein, *et al. Nat. Photonics* (2015)

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J. Wätzel *et al. Opt. Express* (2020)

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A. de las Heras, *et al. Optica* (2022)

M. Luttmann, *et al. Sci. Adv.* (2023)

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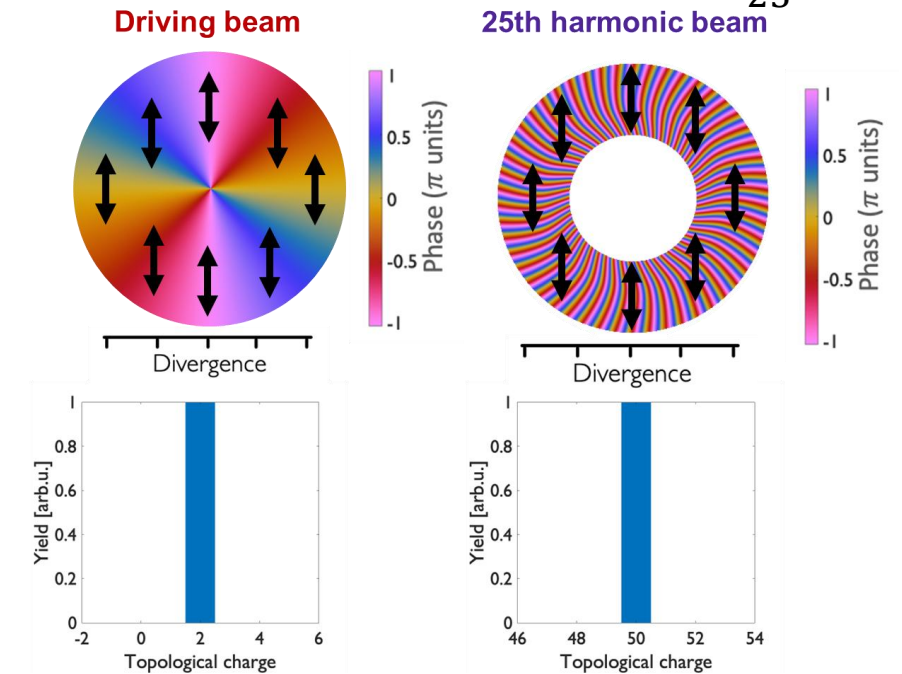
Progress on structured high harmonic beams

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C. Hernández-García *et al.* *Phys. Rev. Lett.* (2013)

OAM selection rule in HHG:

$$\ell = 2 \quad \ell_q = q\ell \quad \ell_{25} = 50$$



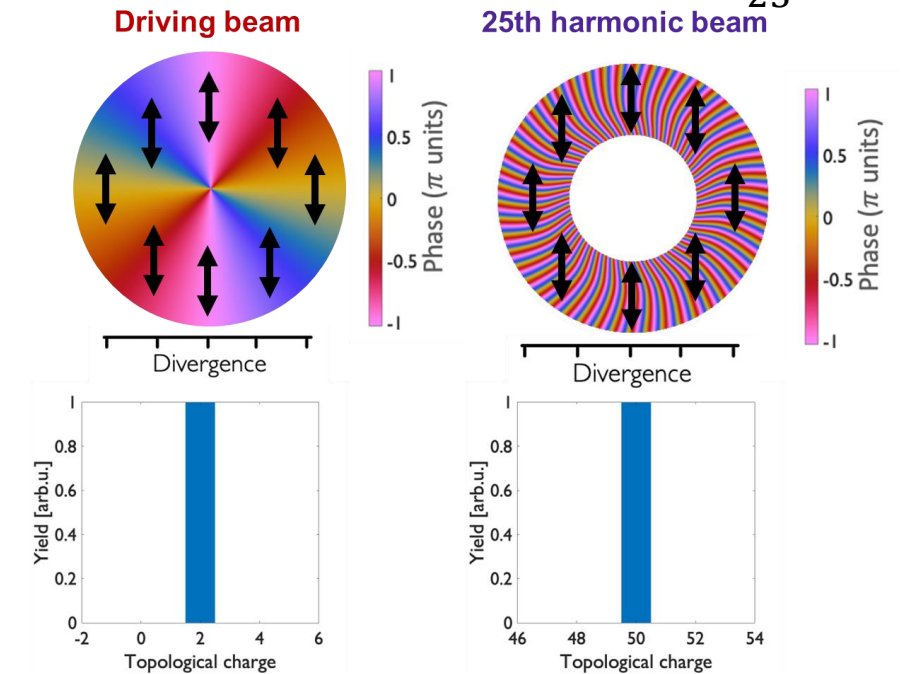
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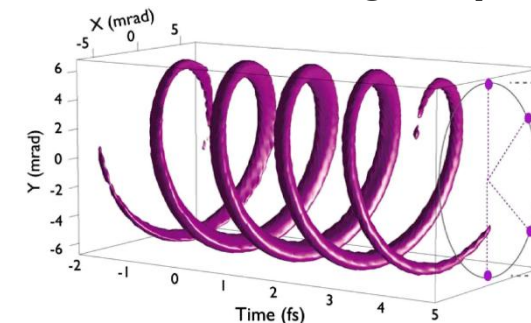
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Attosecond light spring:



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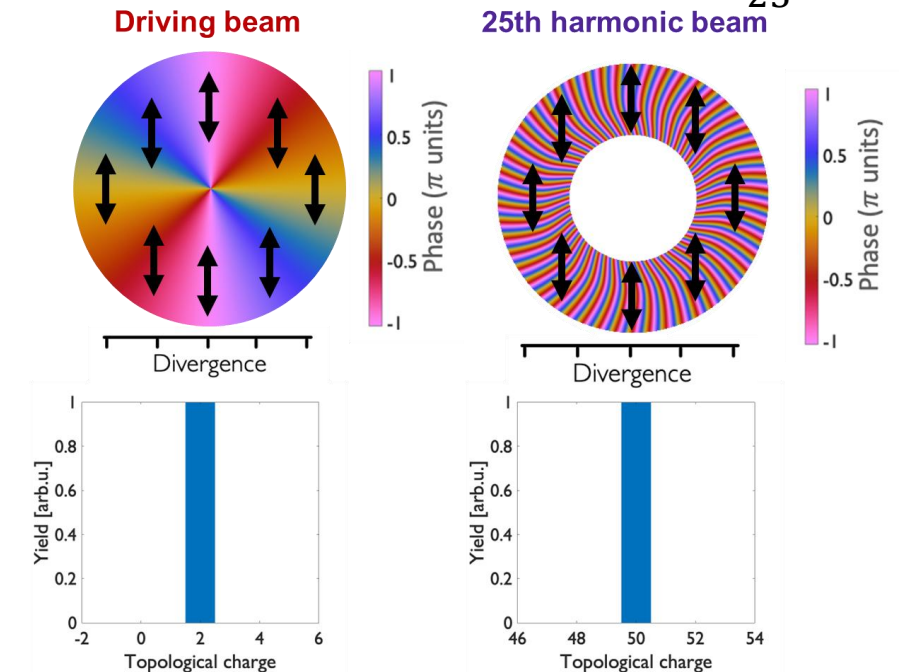
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G. Gariépy *et al. Phys. Rev. Lett.* (2014)

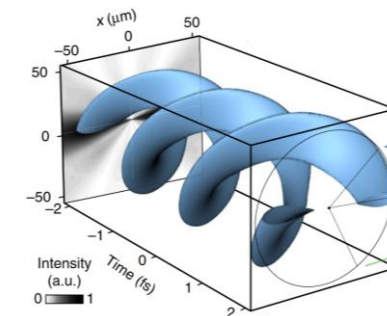
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[A. K. Pandey *et al., ACS Photonics* \(2022\)](#)

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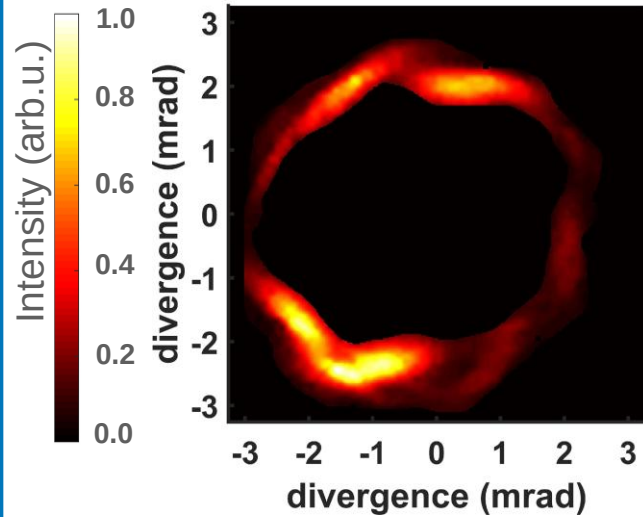
L. Rego *et al. Sci. Adv.* (2022)

Time-dependent OAM (self-torque) in HHG

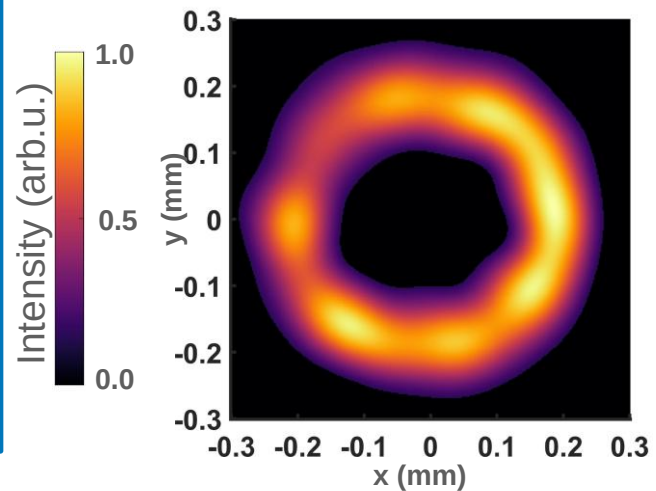
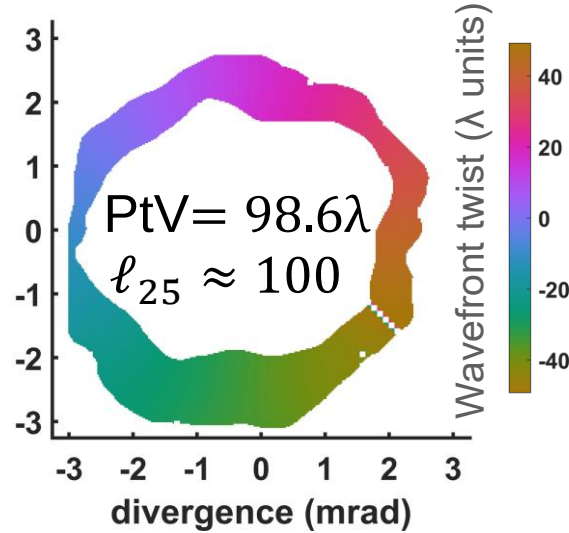
L. Rego *et al. Science* (2019)

Up-conversion of an experimental scalar vortex beam

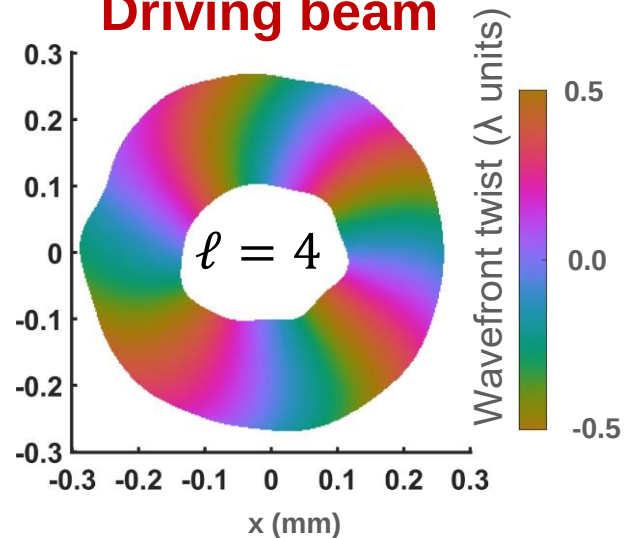
Experiment



25th harmonic beam

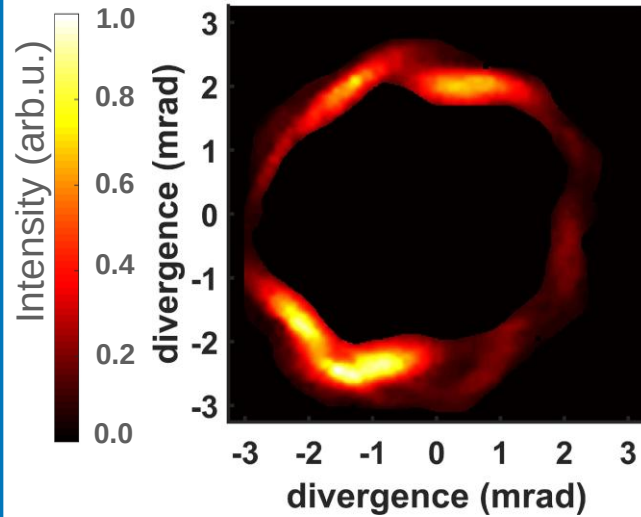


Driving beam

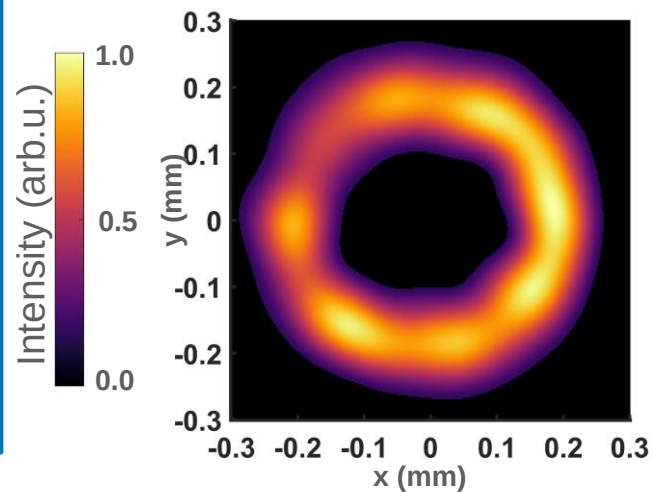
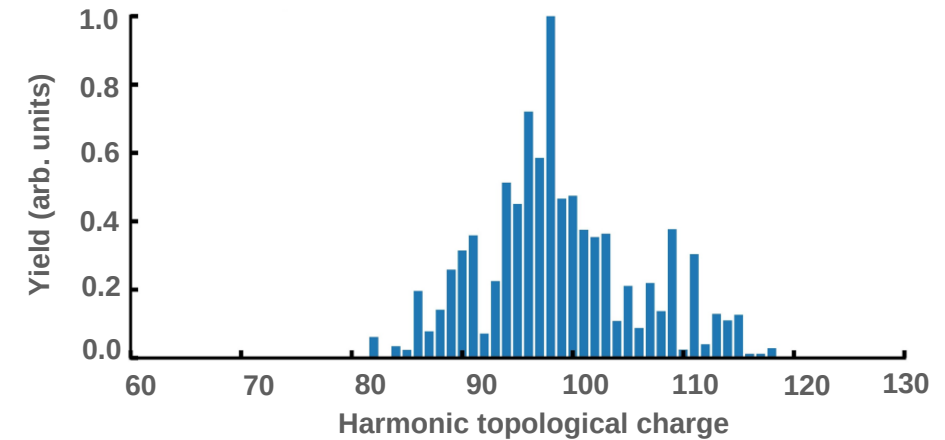
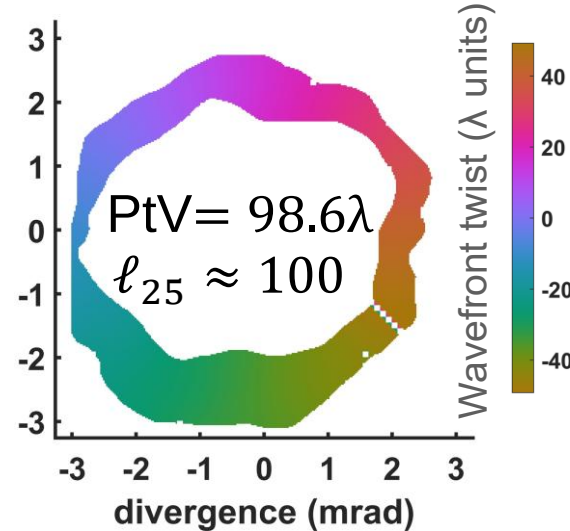


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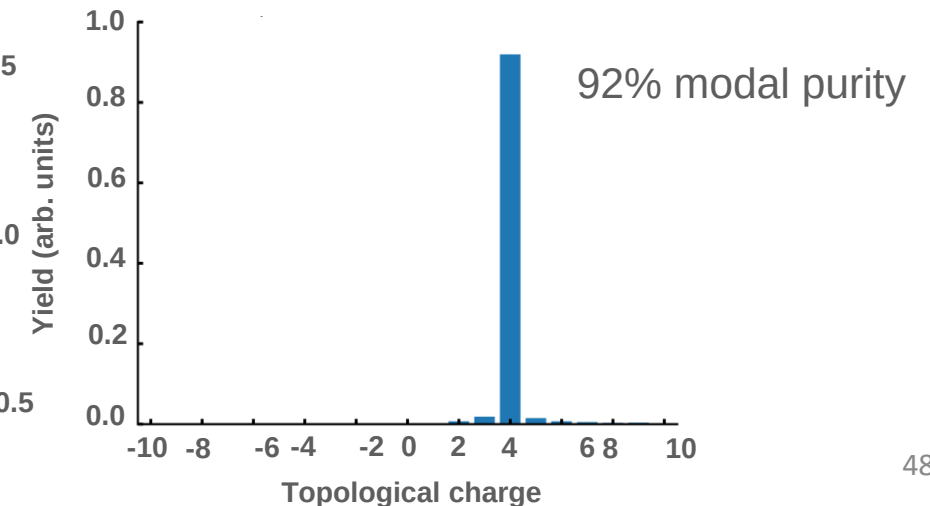
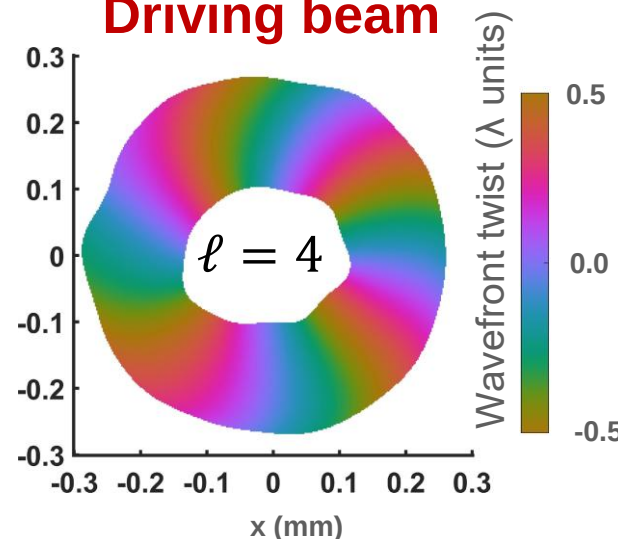
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25th harmonic beam

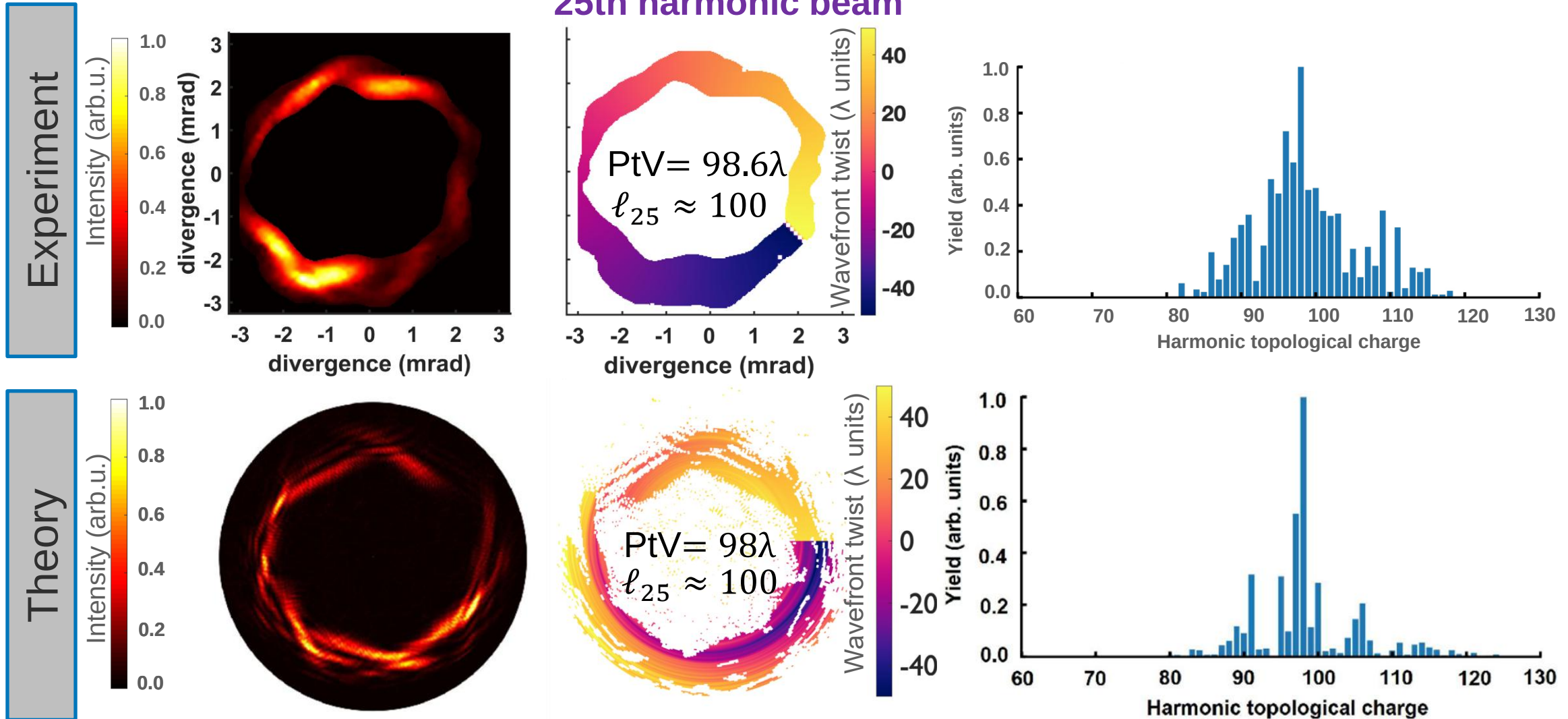


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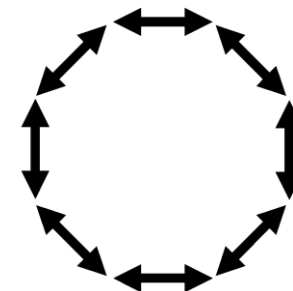
L. Rego *et al.* *Science* (2019)

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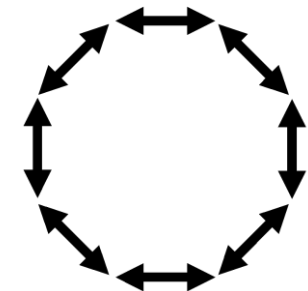
C. Hernández-García *et al.* *Optica* (2017)

J. Wätzel *et al.* *Opt. Express* (2020)

Driving beam



High-harmonic beam



Progress on structured high harmonic beams

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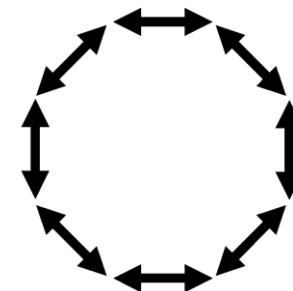
Is it possible to up-convert
vector-vortex beams?

High harmonic up-conversion of vector beams

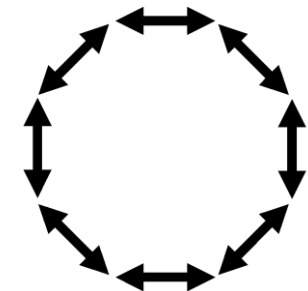
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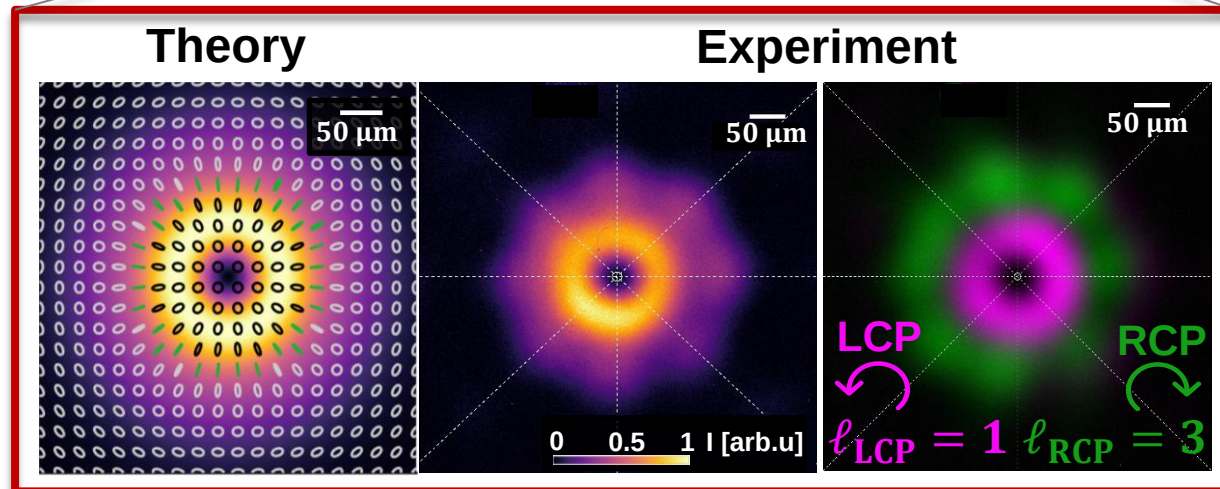
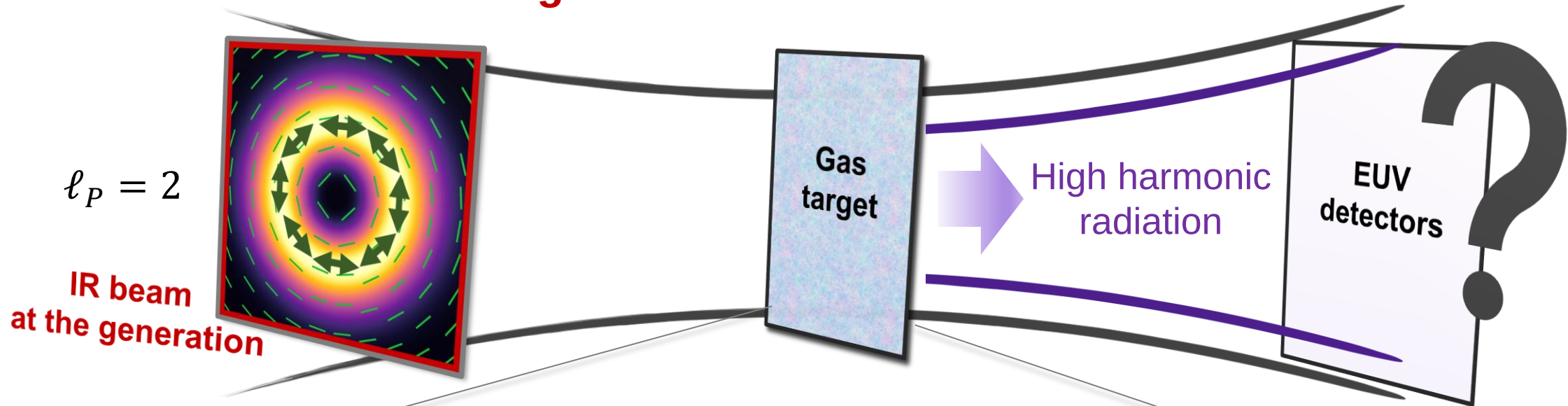


High-harmonic beam



Up-conversion of a **vector vortex beam**

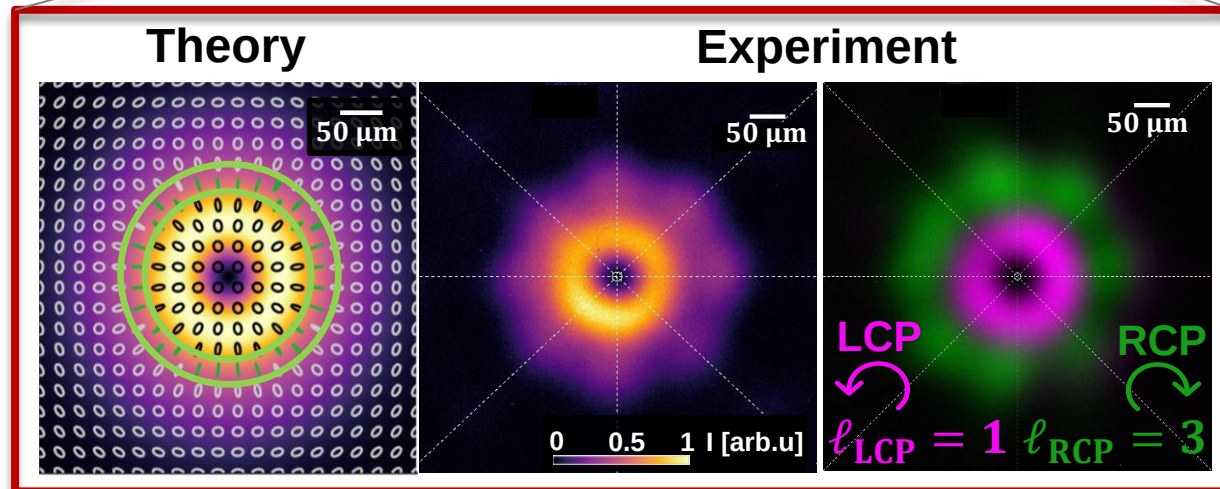
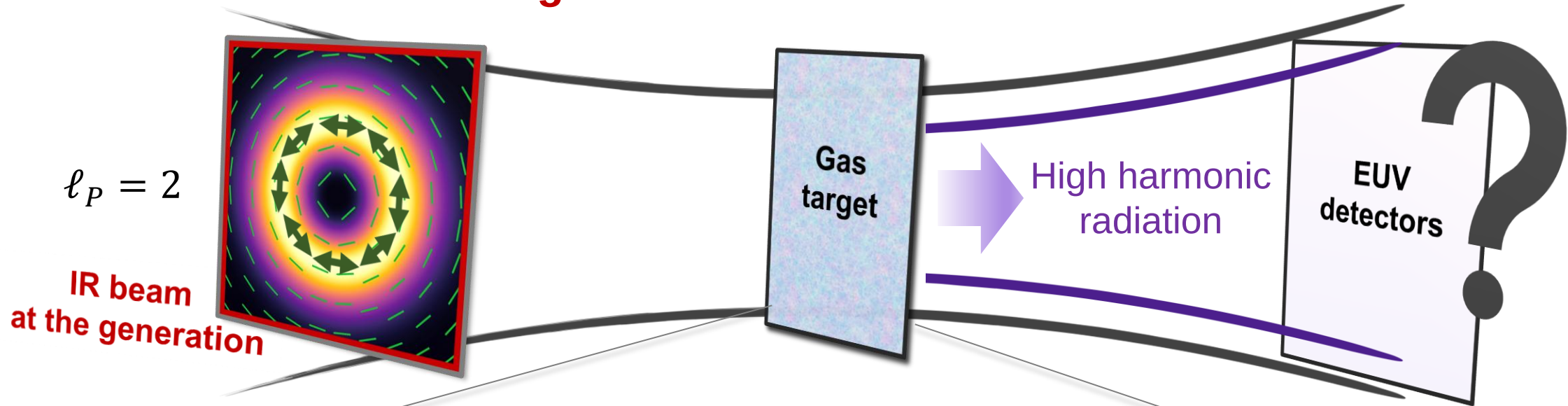
Driving beam



IR beam
at the gas target

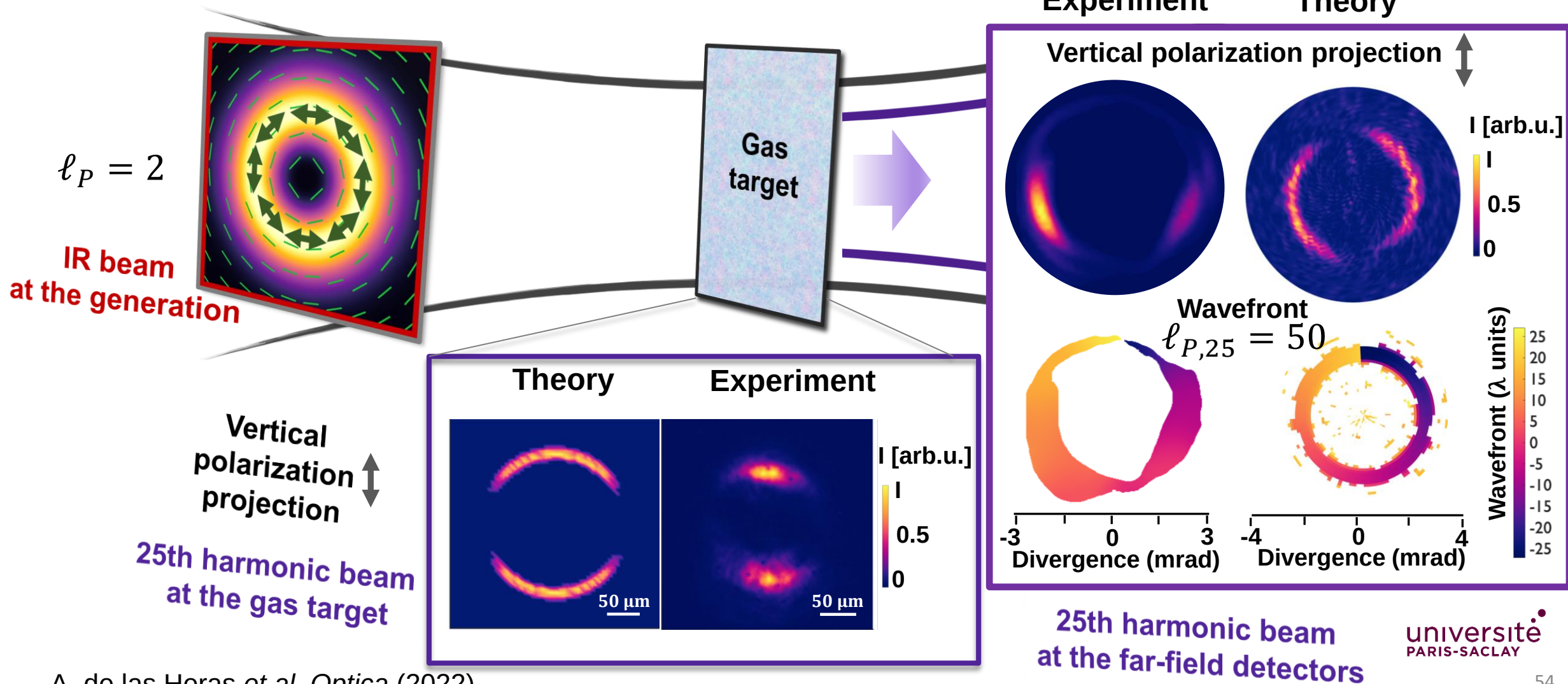
Up-conversion of a **vector vortex beam**

Driving beam

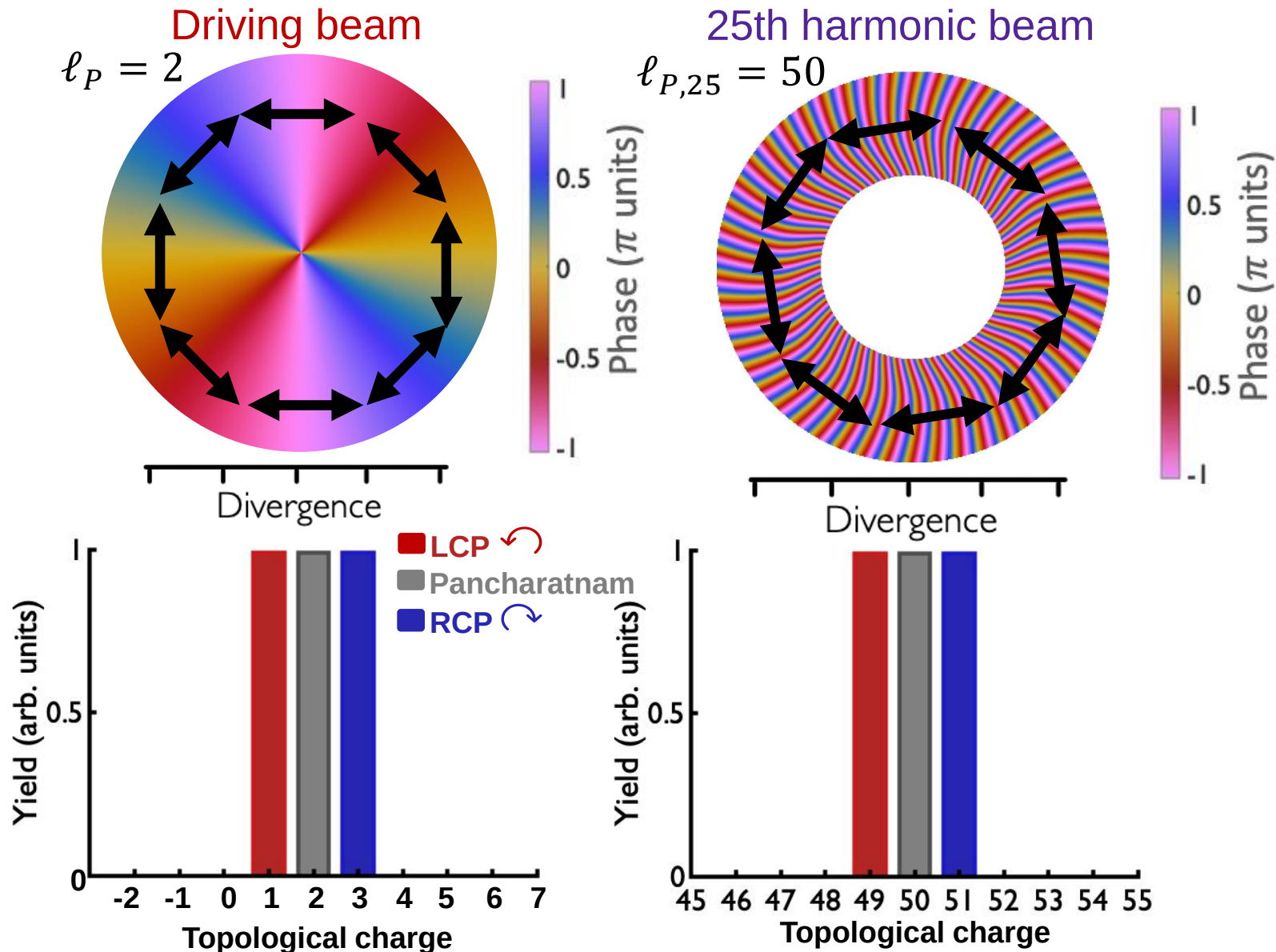


IR beam
at the gas target

Up-conversion of a **vector vortex beam**



Up-conversion of a **vector vortex beam**

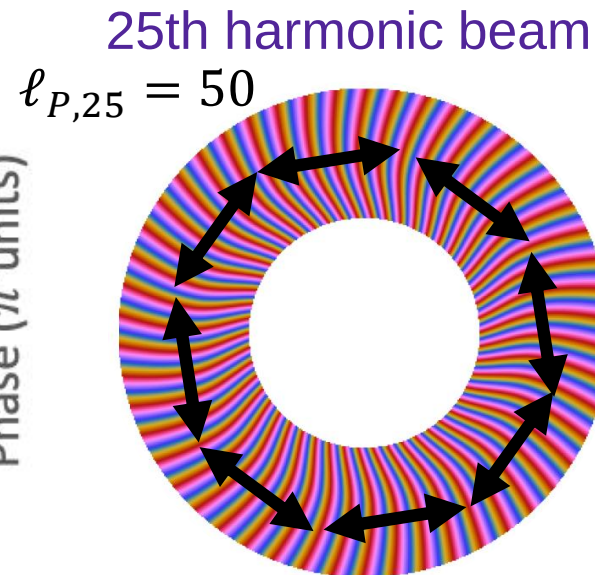
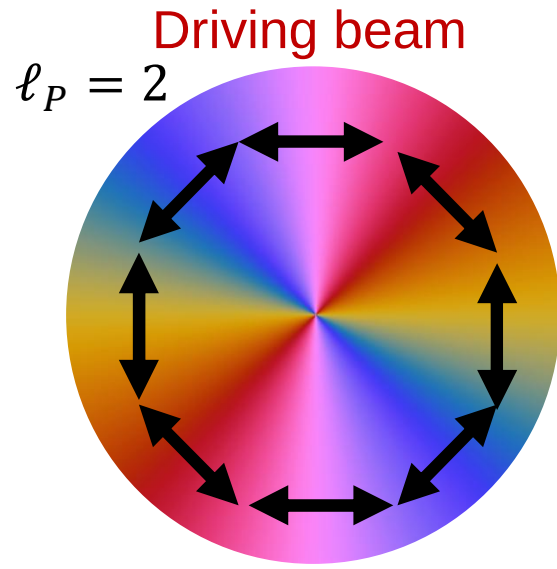


Selection rule in HHG:

$$\ell_{P,q} = q\ell_P$$

A. de las Heras *et al. Optica* (2022)

Up-conversion of a **vector vortex beam**

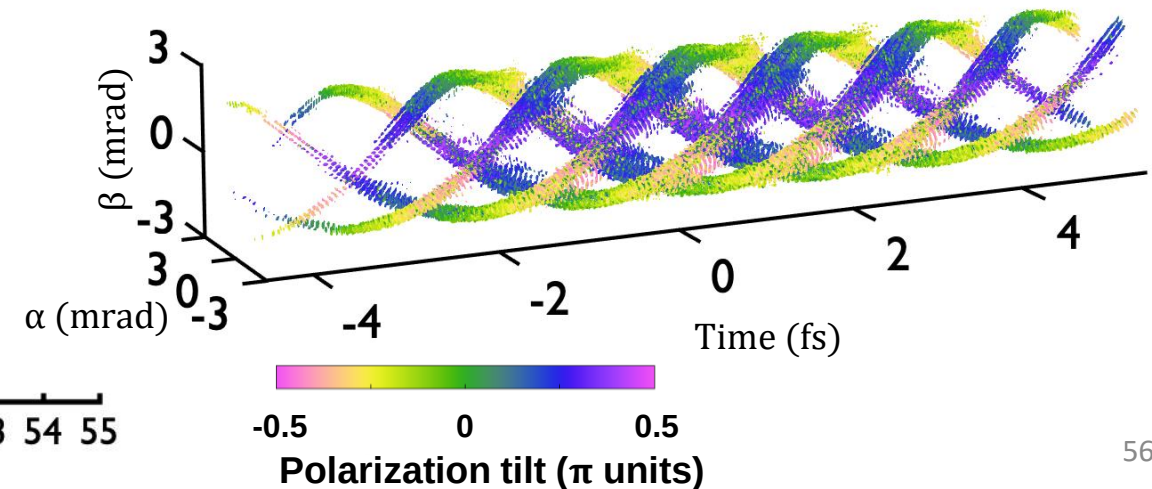
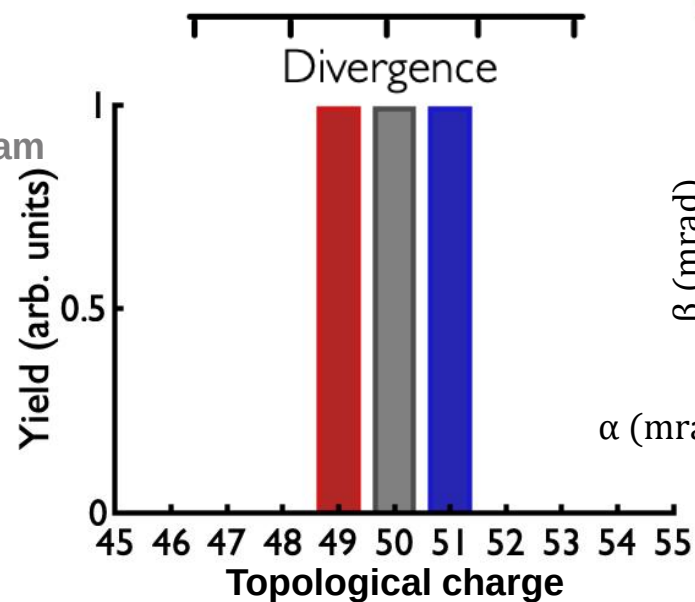
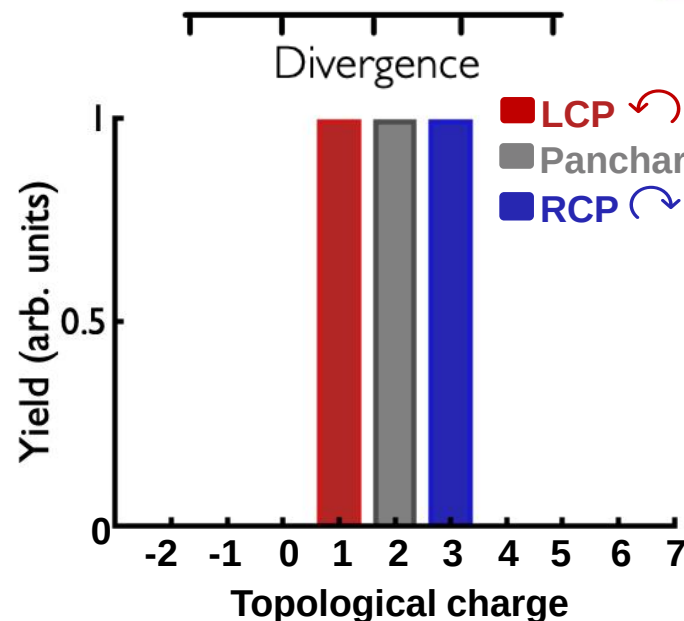


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Attosecond light spring



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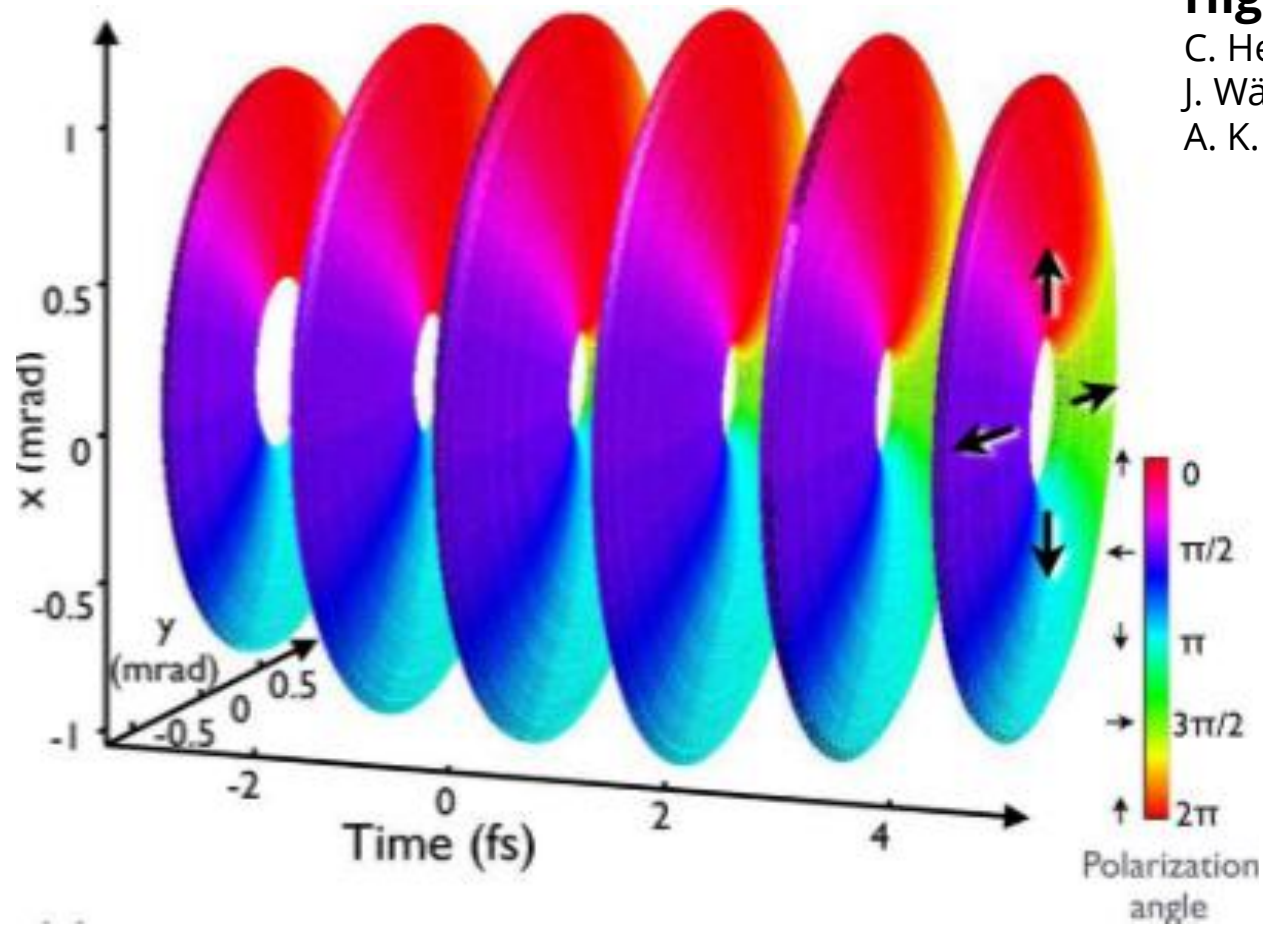
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Self-torque and circular polarization in HHG

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Generation of vectorial attosecond pulse trains



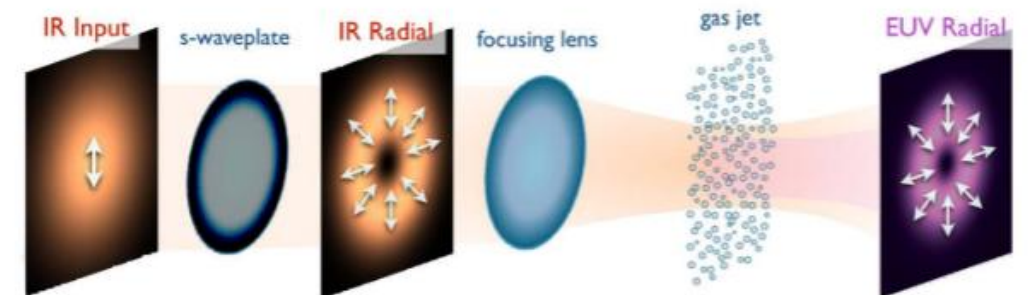
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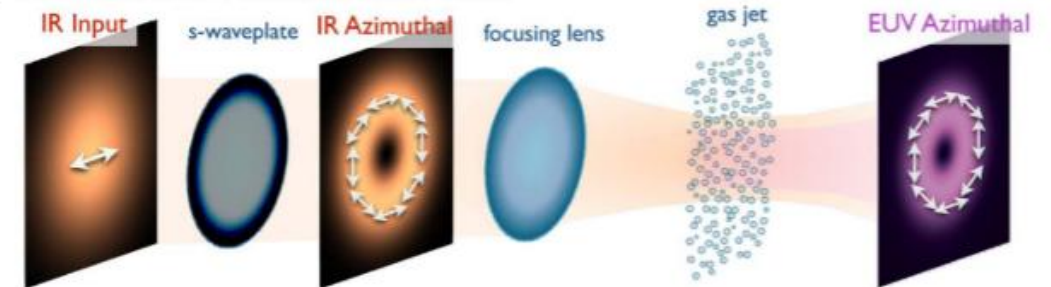
J. Wätzel *et al.* *Opt. Express* (2020)

A. K. Pandey *et al.* *EPJST* (2022)

(a) Generation of EUV radial vector beams



(b) Generation of EUV azimuthal vector beams



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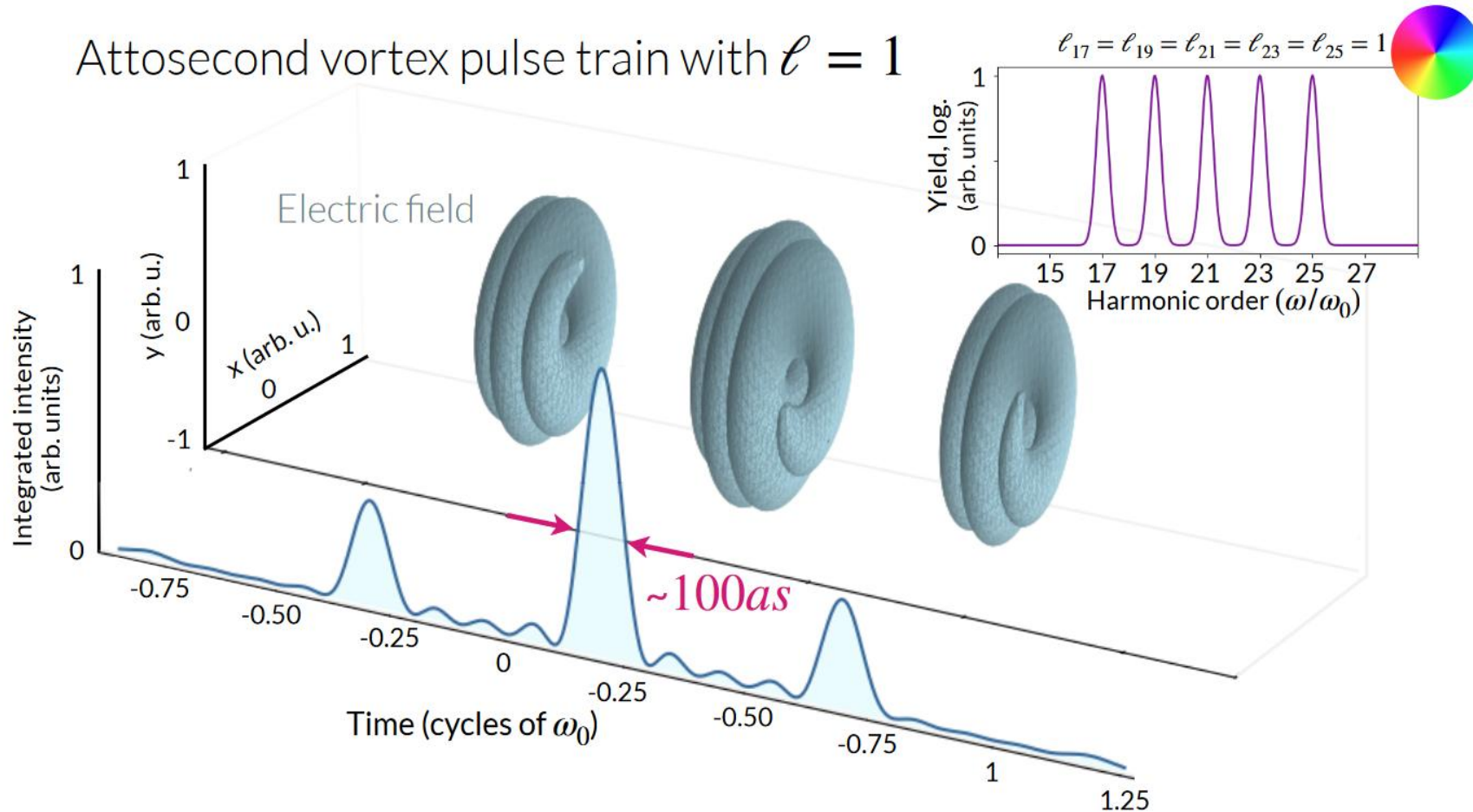
M. Luttmann, *et al. Sci. Adv.* (2023)

N. Brooks *et al. ACS Photonics* (2025)

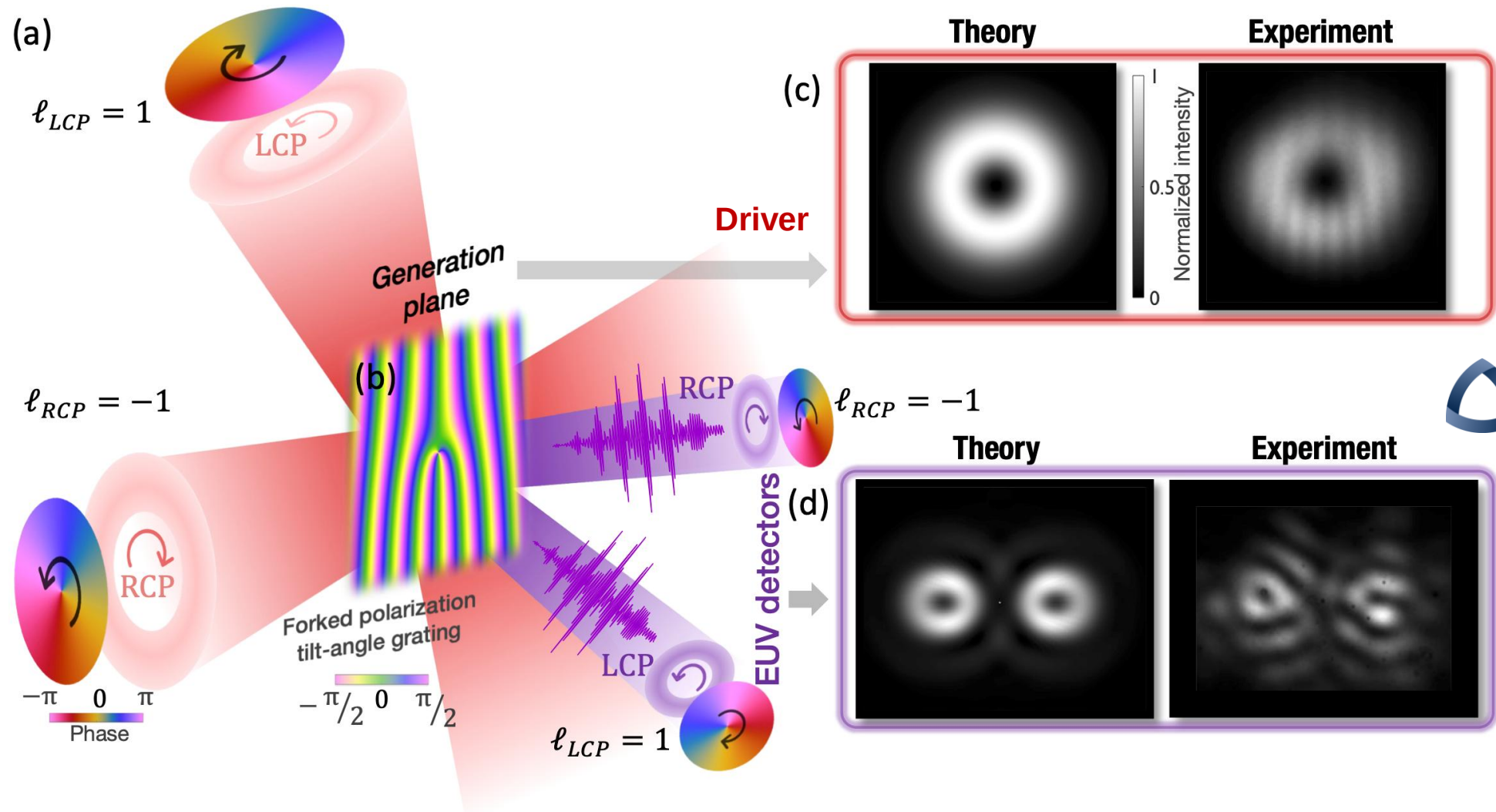
Self-torque and circular polarization in HHG

A. de las Heras *et al. ACS Photonics* (2024)

Generation of attosecond vortex pulses



Generation of attosecond vortex pulse trains from a forked polarization tilt-angle grating driving field



Order-independent OAM in HHG

$$q = n_1 + n_2$$

$$l_q = n_1 l_1 + n_2 l_2 \stackrel{?}{=} (n_1 - n_2) l_1$$

$$\sigma_q = n_1 \sigma_1 + n_2 \sigma_2 \stackrel{?}{=} (n_1 - n_2) \sigma_1$$

$$\sigma_2 = -\sigma_1$$

Two possible channels:

$$\sigma_q = \sigma_1, n_2 = n_1 - 1, q = 2n_1 - 1, l_q = l_1$$

$$\sigma_q = -\sigma_1, n_2 = n_1 + 1, q = 2n_1 + 1, l_q = -l_1$$

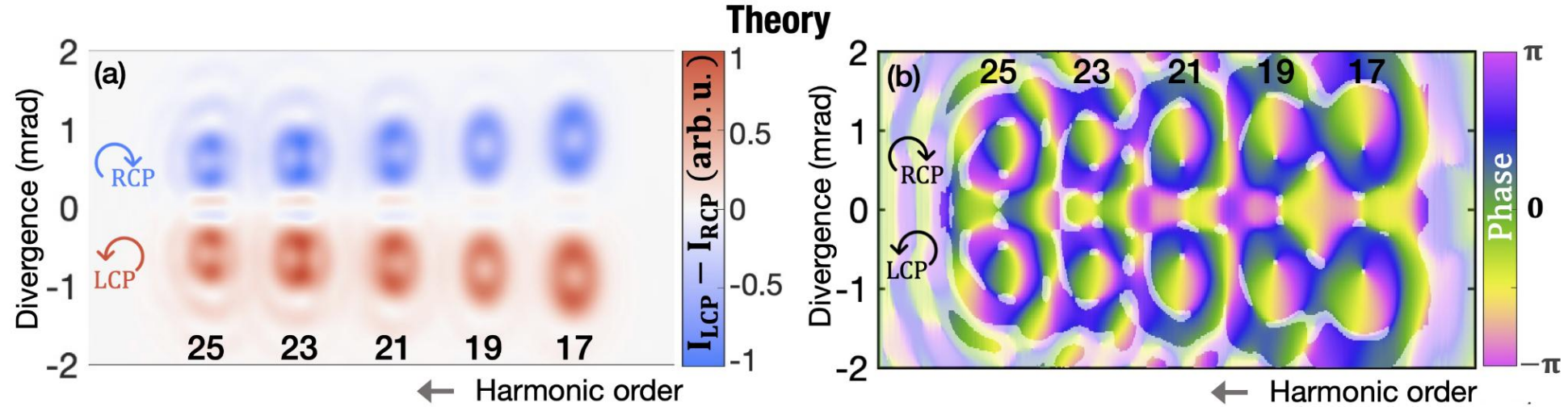
$$\text{Example: } n_1 = 10, n_2 = 9 \rightarrow \sigma_q = \sigma_1, q = 19, l_q = l_1$$

$$n_1 = 9, n_2 = 10 \rightarrow \sigma_q = -\sigma_1, q = 19, l_q = -l_1$$

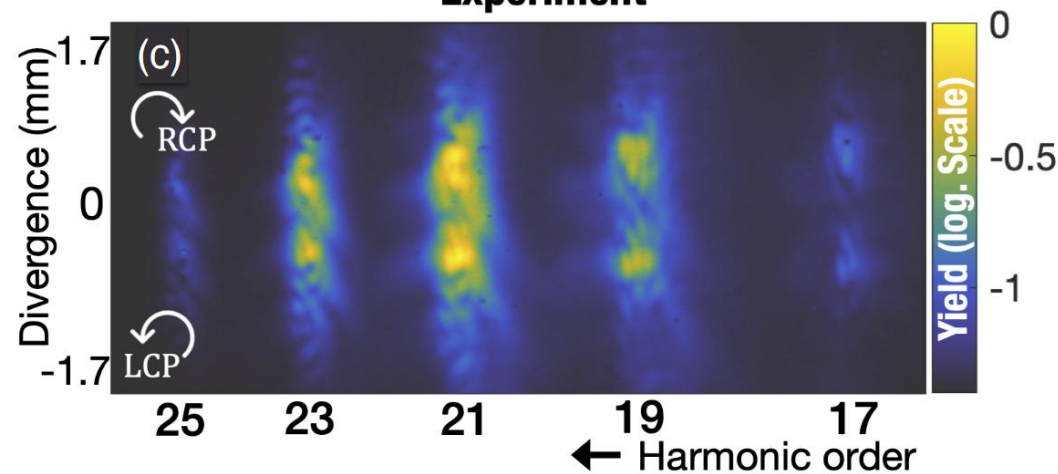
$$\vec{k}_q = n_1 \vec{k}_1 + n_2 \vec{k}_2, \vec{k}_1 \neq \vec{k}_2$$

Generation of attosecond vortex pulse trains from a forked polarization tilt-angle grating driving field

Far-field high harmonic spectrum

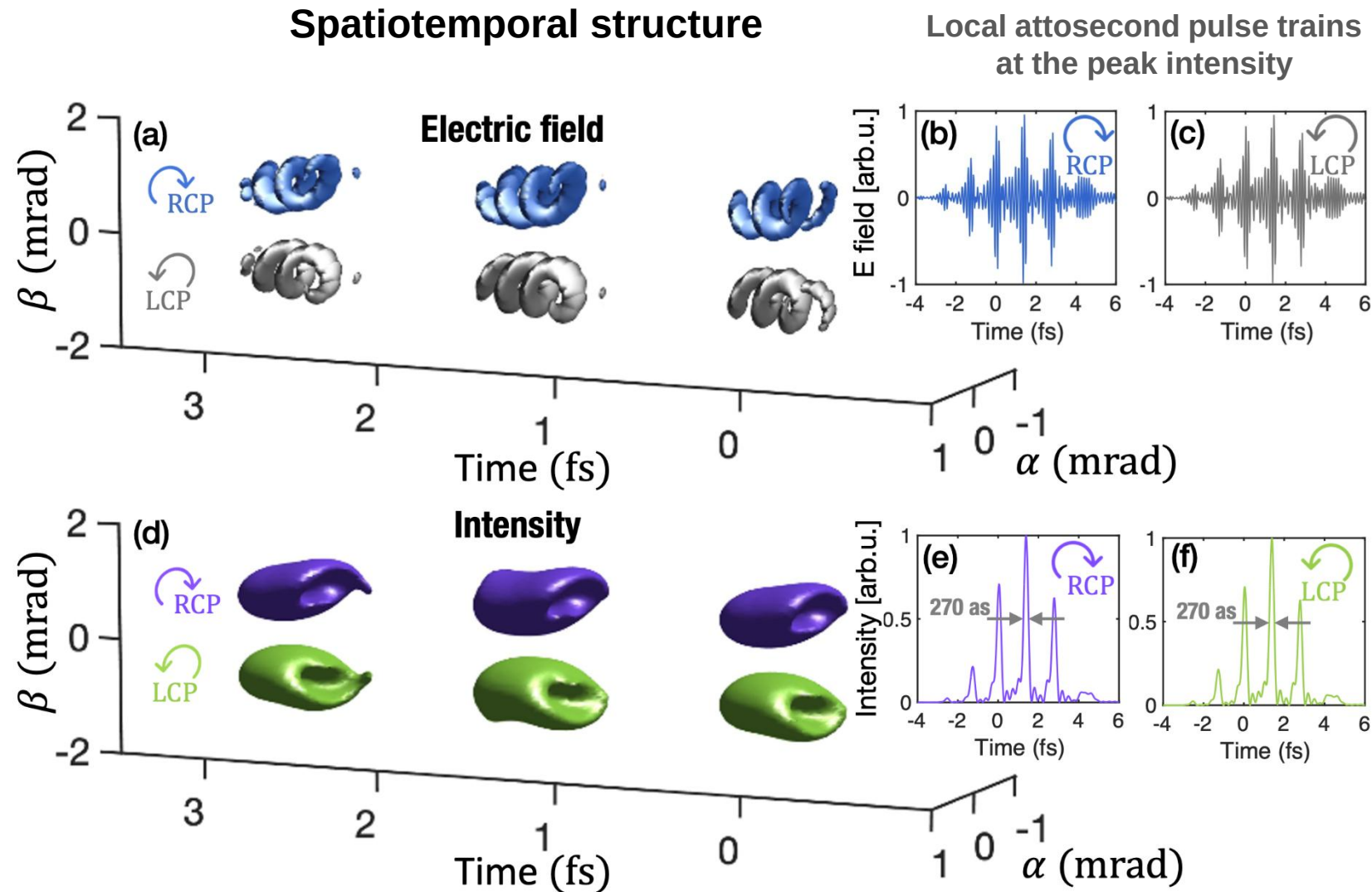


Experiment



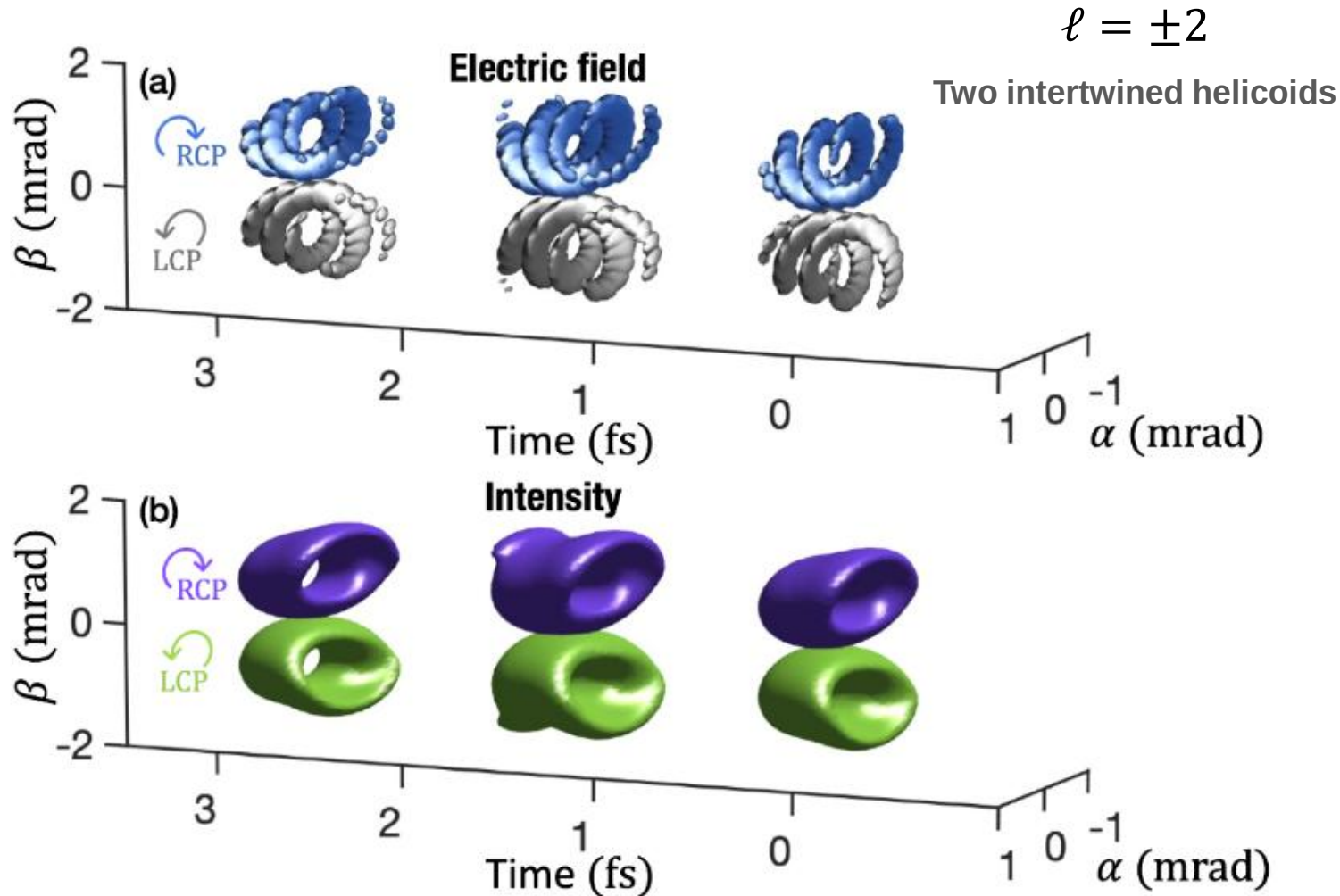
Generation of attosecond vortex pulse trains from a forked polarization tilt-angle grating driving field

$$\ell = \pm 1$$

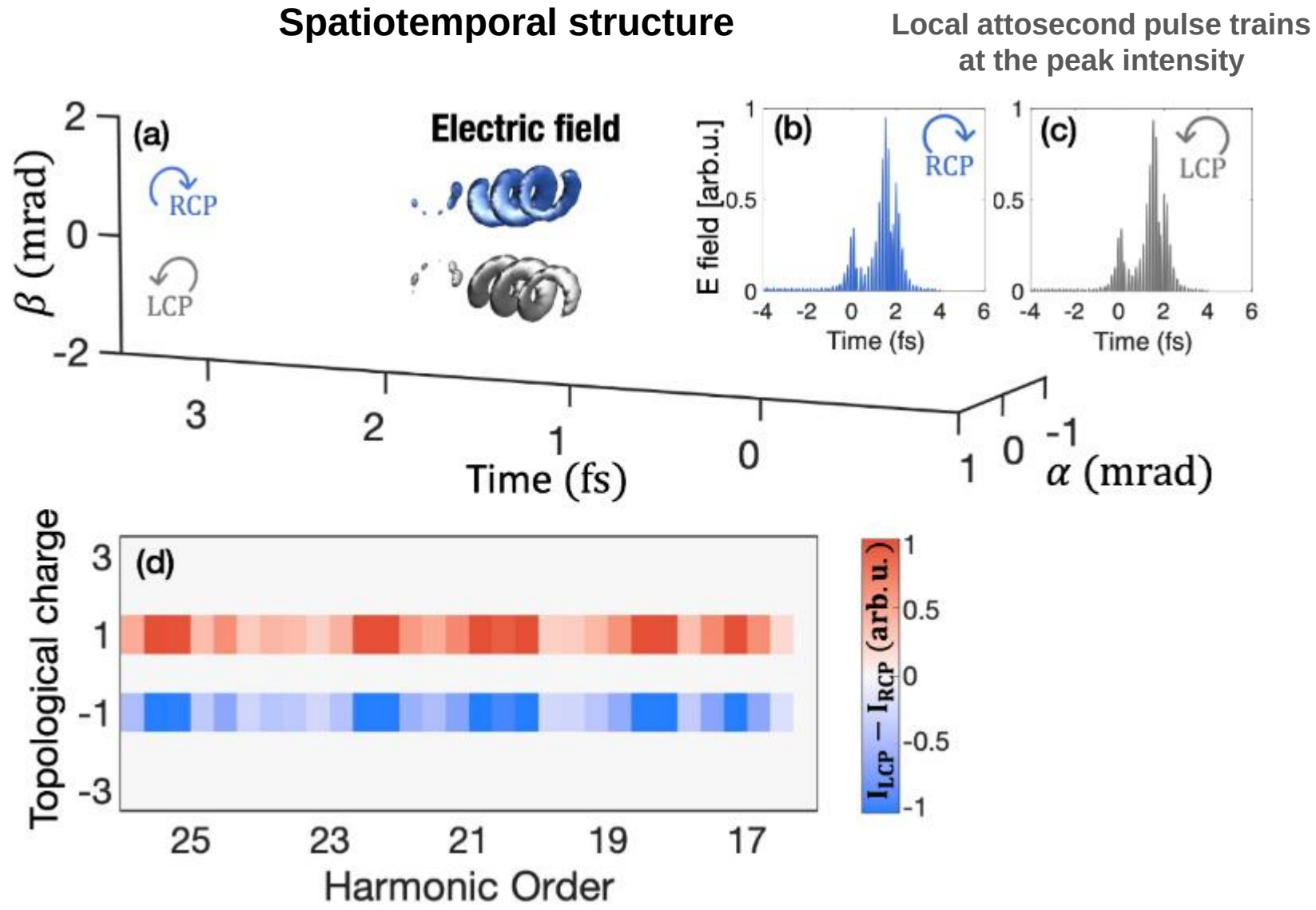


Generation of attosecond vortex pulse trains from a forked polarization tilt-angle grating driving field

Spatiotemporal structure

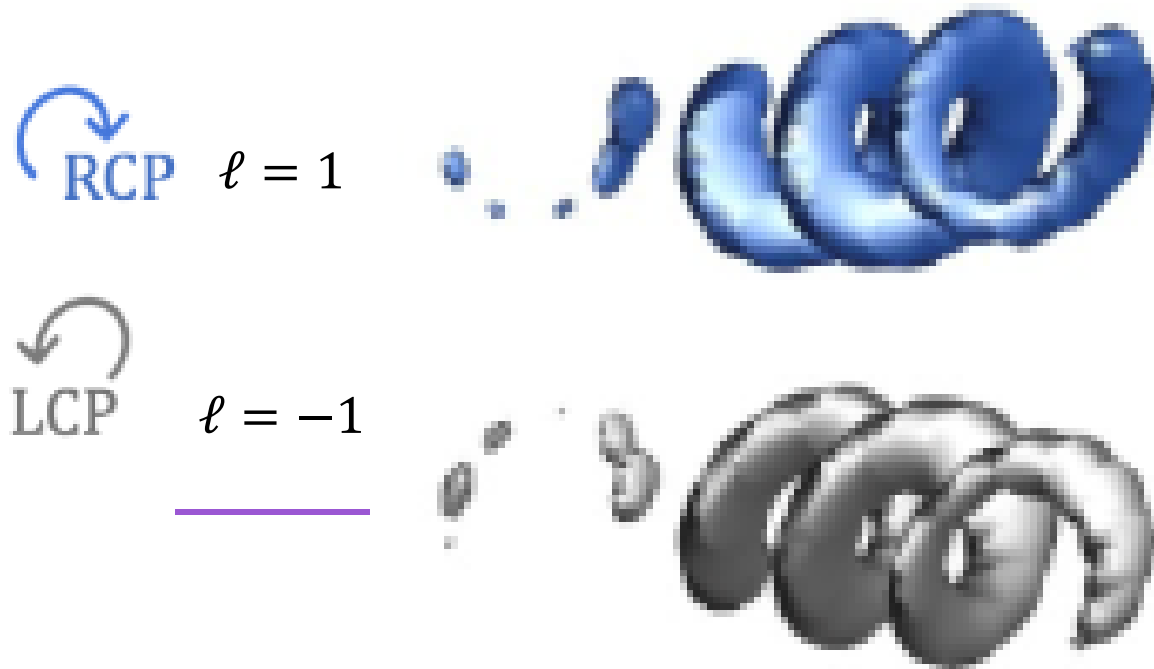


Generation of a single attosecond vortex pulse from a forked polarization tilt-angle grating driving field



Merging helical and circular dichroism for chiral detection

Future perspective: exploiting spin and orbital angular momentum in attosecond pulses or high-frequency beams for combined helical and circular dichroism.



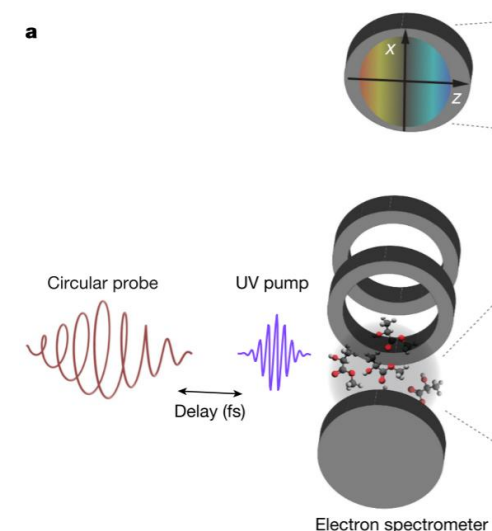
Time-resolved circular dichroism

Article | [Open access](#) | Published: 22 May 2024

Capturing electron-driven chiral dynamics in UV-excited molecules

[Vincent Wanie](#) , [Etienne Bloch](#), [Erik P. Månsson](#), [Lorenzo Colaizzi](#), [Sergey Ryabchuk](#), [Krishna Saraswathula](#), [Andres F. Ordonez](#), [David Ayuso](#), [Olga Smirnova](#), [Andrea Trabattoni](#), [Valérie Blanchet](#), [Nadia Ben Amor](#), [Marie-Catherine Heitz](#), [Yann Mairesse](#), [Bernard Pons](#)  & [Francesca Calegari](#) 

[Nature](#) **630**, 109–115 (2024) | [Cite this article](#)



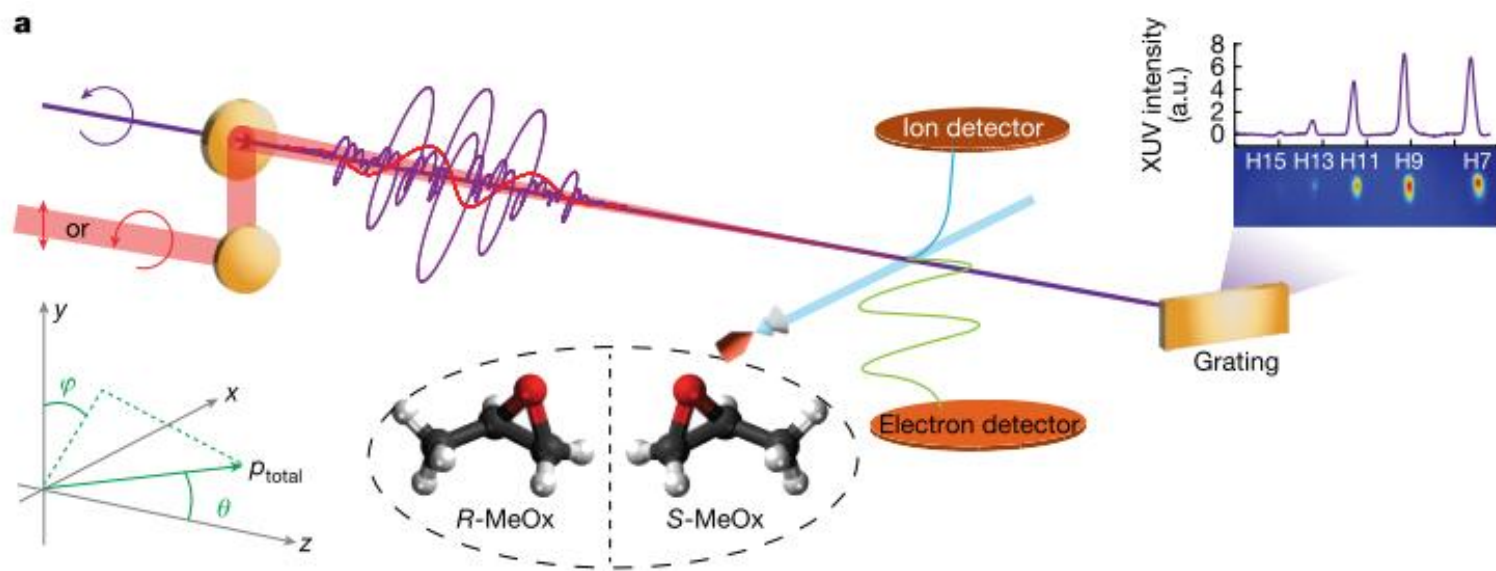
Attosecond circular dichroism

Article | [Open access](#) | Published: 27 August 2025

Attosecond control and measurement of chiral photoionization dynamics

[Meng Han](#) , [Jia-Bao Ji](#), [Alexander Blech](#), [R. Esteban Goetz](#), [Corbin Allison](#), [Loren Greenman](#), [Christiane P. Koch](#) & [Hans Jakob Wörner](#) 

[Nature](#) **645**, 95–100 (2025)



High-frequency helical dichroism

Article | Published: 04 July 2022

Hard X-ray helical dichroism of disordered molecular media

Jérémy R. Rouxel , Benedikt Rösner , Dmitry Karpov, Camila Bacellar, Giulia F. Mancini, Francesco Zinna, Dominik Kinschel, Oliviero Cannelli, Malte Oppermann, Cris Svetina, Ana Diaz, Jérôme Lacour, Christian David & Majed Chergui 

Nature Photonics **16**, 570–574 (2022) | [Cite this article](#)

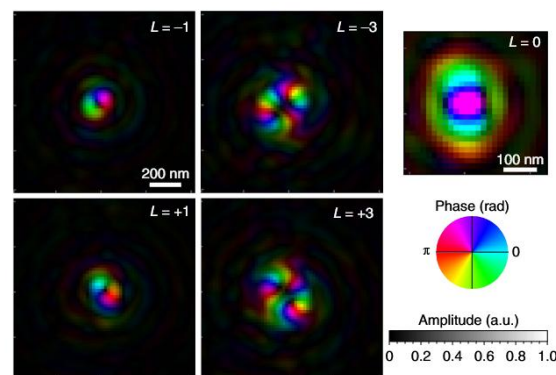
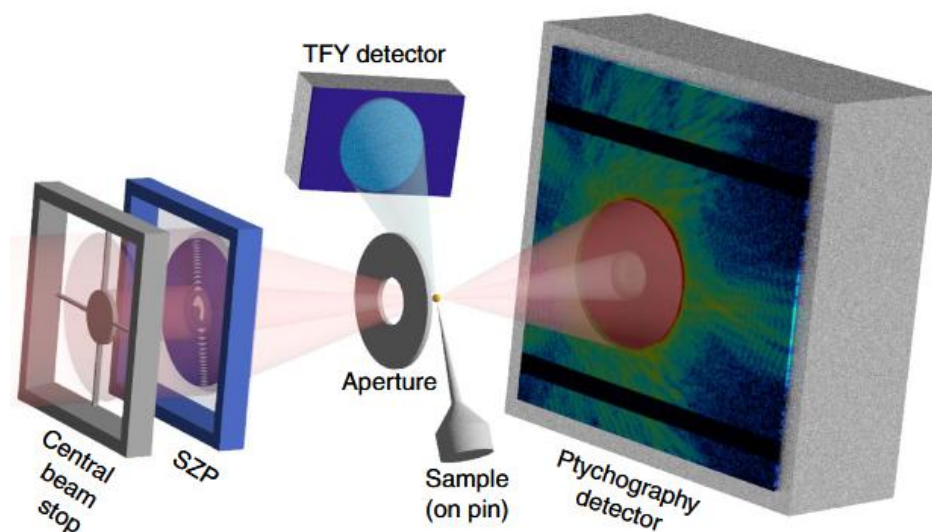


Fig. 3 | Ptychographic reconstructions of helical beams used in the subsequent HD measurements. The OAM of the beam is indicated by the L value in each panel. The colour wheel and the greyscale bar indicate, respectively, the phase and amplitude of the field at the focal point.

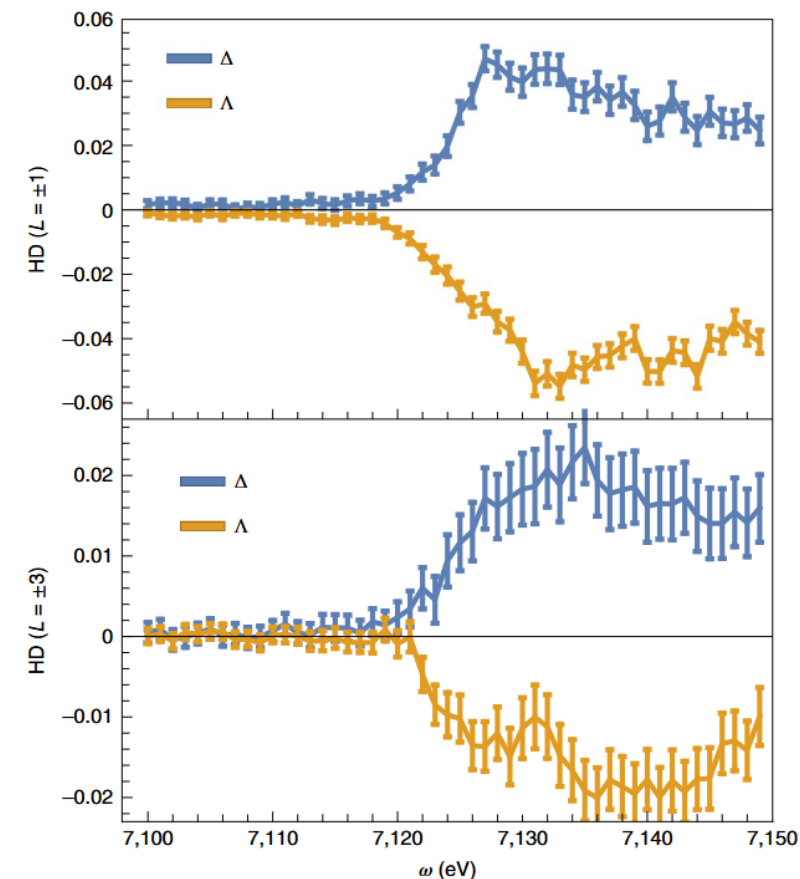


Fig. 5 | HD signals at the Fe K edge of Δ - and Λ -[Fe(4,4'-diMebpy)₃] salts for $L = \pm 1$ (top) and $L = \pm 3$ (bottom). Error bars represent the mean $\pm$ the combined standard uncertainty (see Supplementary Section 6).

Imprinting OAM into propagating electrons

Letter | Published: 10 August 2020

Photoelectric effect with a twist

[Giovanni De Ninno](#) , [Jonas Wätzel](#), [Primož Rebernik Ribič](#), [Enrico Allaria](#), [Marcello Coreno](#), [Miltcho B. Danailov](#), [Christian David](#), [Alexander Demidovich](#), [Michele Di Fraia](#), [Luca Giannessi](#), [Klavs Hansen](#), [Špela Krušič](#), [Michele Manfreda](#), [Michael Meyer](#), [Andrej Mihelič](#), [Najmeh Mirian](#), [Oksana Plekan](#), [Barbara Ressel](#), [Benedikt Rösner](#), [Alberto Simoncig](#), [Simone Spampinati](#), [Matija Stupar](#), [Matjaž Žitnik](#), [Marco Zangrando](#), [Carlo Callegari](#) & [Jamal Berakdar](#) 

[Nature Photonics](#) **14**, 554–558 (2020) | [Cite this article](#)

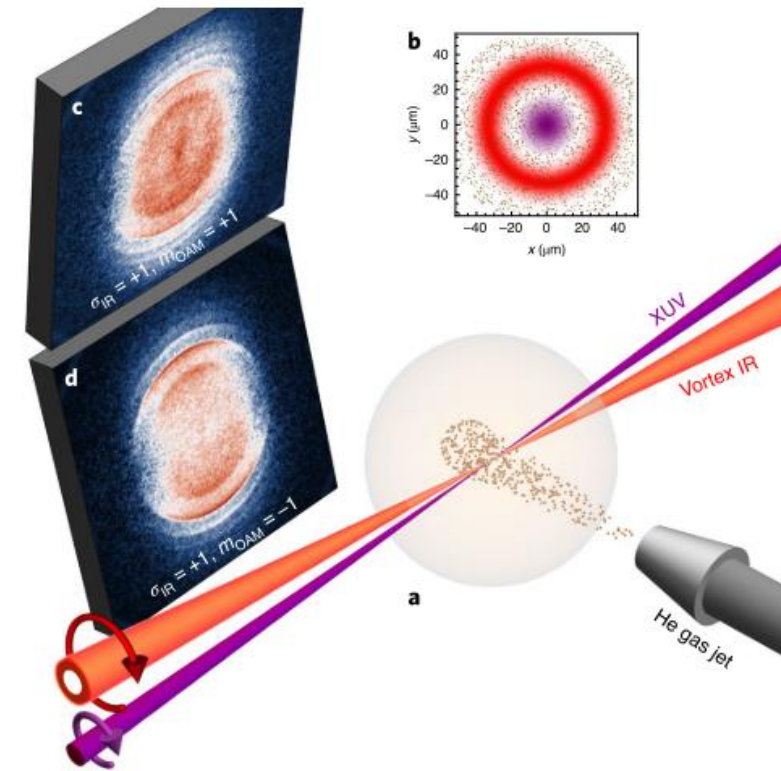
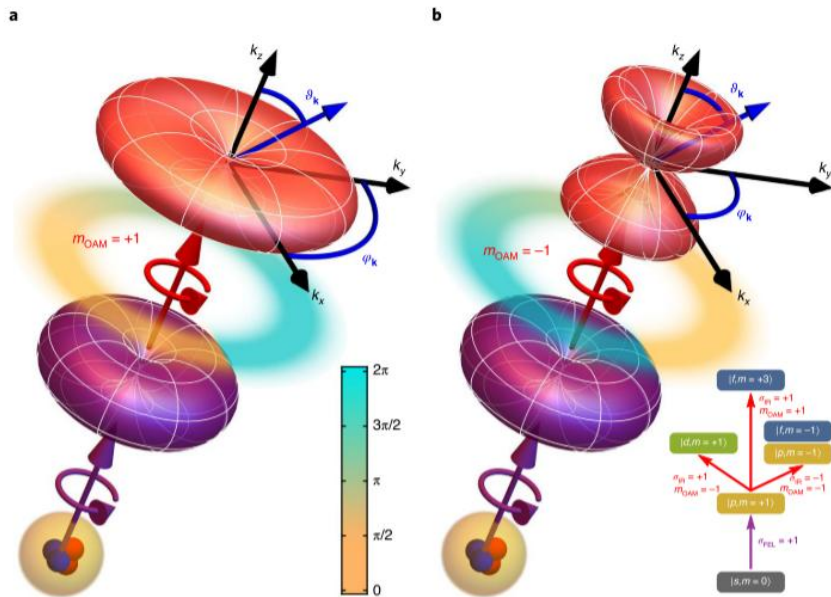


Fig. 1 | Experimental set-up and first evidence of OAM-dependent dichroism. a, Schematic representation of the experimental set-up. The XUV light pulse is generated using a FEL. It has a fixed (right-handed) circular polarization, a spot size of $\sim 10 \mu\text{m}$ and power density (at the sample position) of the order of 10^{18} W m^{-2} . The infrared laser beam has a spot size of $\sim 60 \mu\text{m}$ and a power density in the range $0.5\text{--}4 \times 10^{17} \text{ W m}^{-2}$. It provides variable circular (right- and left-handed) polarization and can be made to carry a topological charge ($m_{\text{OAM}} = \pm 1$). Helium gas is injected into the experimental chamber (at room temperature) via a gas jet system. A VMI spectrometer is used to determine the angular distribution of the signal generated by the photoemitted electrons. **b**, Real-size XUV-FEL (blue) and infrared (red) spatial profiles at the waist position. Helium atoms are also represented as dots with exaggerated dimensions. **c**, VMI image obtained for high infrared intensity ($\sim 3 \times 10^{17} \text{ W m}^{-2}$), right-handed infrared polarization and positive topological charge ($m_{\text{OAM}} = +1$). **d**, As in **c**, but with negative topological charge ($m_{\text{OAM}} = -1$).

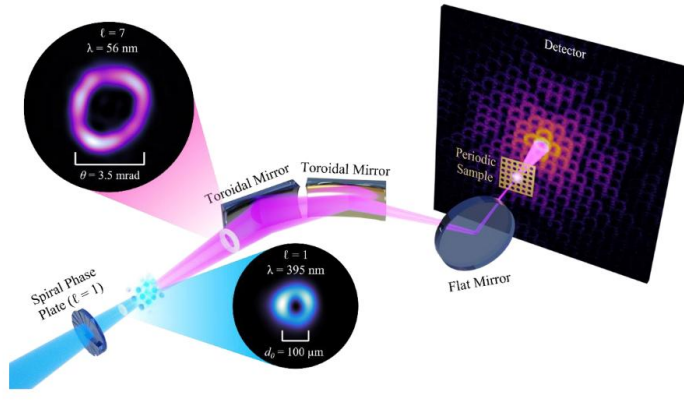
Structured light allows for

- Further capabilities in different applications

Optical trapping

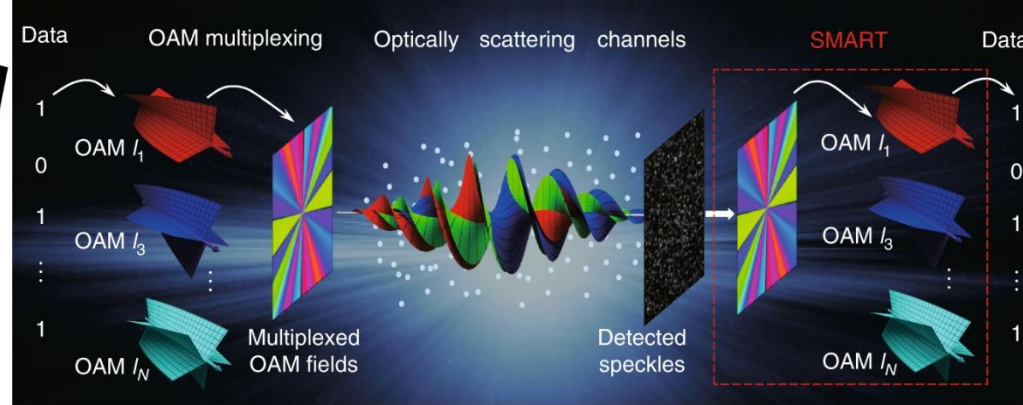


Imaging



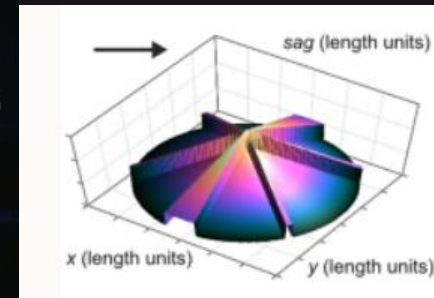
B. Wang et al. *Optica* (2023)

Multiplexing



L. Gong et al. *LSA* (2019)

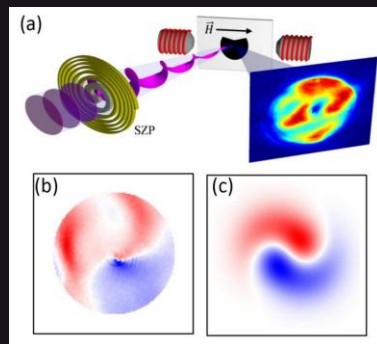
Material processing



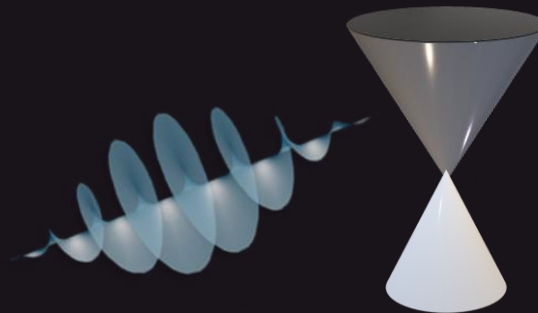
D. Flamm et al. *Opt. Engineering* (2021)

Y. Yang et al. *Adv. Photonics* (2021)

- Novel scenarios in the interplay of light and matter symmetries



M. Fanciulli et al. *PRL* (2021)



Autoevaluation form of SESSION 3

<https://forms.gle/4pmq7QLmRhSspno86>

